Objective of the case study

- Perform descriptive analytics to create a customer profile for each AeroFit treadmill product by developing appropriate tables and charts.
- For each AeroFit treadmill product, construct two-way contingency tables and compute all conditional and marginal probabilities along with their insights/impact on the business.

In [53]:
```python
# Importing Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import math
from scipy.stats import binom,poisson,norm,expon,geom,uniform
```

In [3]:
```python
# Importing Dataset

df = pd.read_csv('aerofit_data.csv')
df.head()
```

Out[3]:

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

In [ ]:
```python
# Checking the dataset and understanding the datatypes, size and null values
```

In [9]:
```python
print('The sie of the dataframe is:',df.shape)
```

```
The sie of the dataframe is: (180, 9)
```

In [4]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Product        180 non-null    object
 1   Age            180 non-null    int64
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64
 6   Fitness        180 non-null    int64
 7   Income         180 non-null    int64
 8   Miles          180 non-null    int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

In [10]:
```python
df['Product'].value_counts()
```

Out[10]:

Product	count
KP281	80
KP481	60
KP781	40

dtype: int64

Comments:

* Overall we have 180 rows and 9 columns with no Null values

* Since the objective is to create a customer profile based on the data given for each treadmill so the approach would be to take a treadmill, create a customer profile and replicate the same for other treadmills. Will start with the highest selling treadmill

KP281 Product

In [186…
```python
first_product = df[df['Product'] == 'KP281']
first_product.shape # All the data belong to KP281 are filtered
```

Out[186…
```
(80, 9)
```

In [187…
```python
first_product.head()
```

Out[187…

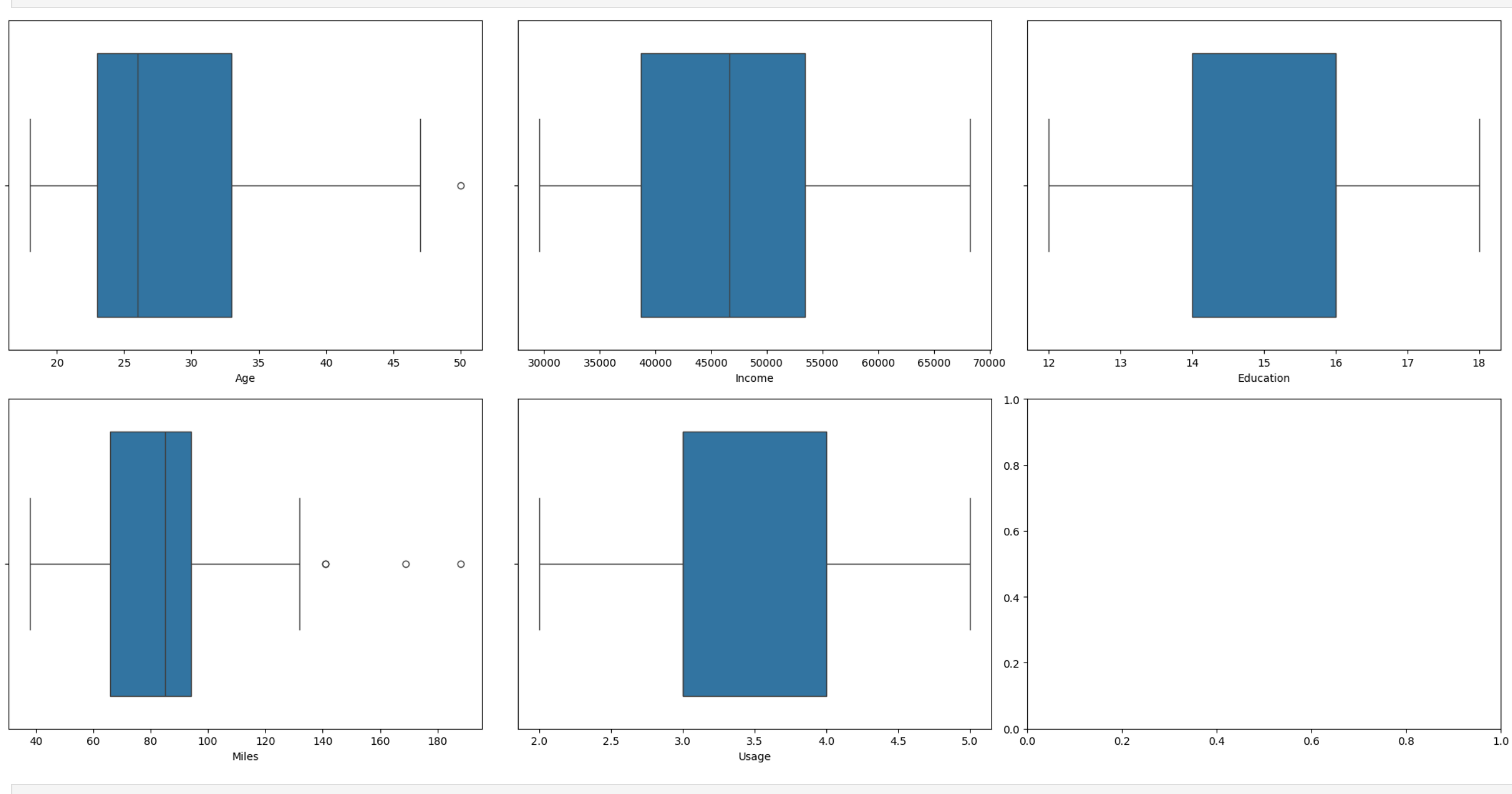
	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

In [23]:
```python
first_product.describe()
```

Out[23]:

	Age	Education	Usage	Fitness	Income	Miles
count	80.000000	80.000000	80.000000	80.00000	80.00000	80.000000
mean	28.550000	15.037500	3.087500	2.96250	46418.02500	82.787500
std	7.221452	1.216383	0.782624	0.66454	9075.78319	28.874102
min	18.000000	12.000000	2.000000	1.00000	29562.00000	38.000000
25%	23.000000	14.000000	3.000000	3.00000	38658.00000	66.000000
50%	26.000000	16.000000	3.000000	3.00000	46617.00000	85.000000
75%	33.000000	16.000000	4.000000	3.00000	53439.00000	94.000000
max	50.000000	18.000000	5.000000	5.00000	68220.00000	188.000000

In [69]:
```python
# First check for outliers for all columns

fig, ax = plt.subplots(2, 3, figsize=(20, 10))

sns.boxplot(x=first_product['Age'], ax=ax[0, 0])
sns.boxplot(x=first_product['Income'], ax=ax[0, 1])
sns.boxplot(x=first_product['Education'], ax=ax[0, 2])
sns.boxplot(x=first_product['Miles'], ax=ax[1, 0])
sns.boxplot(x=first_product['Usage'], ax=ax[1, 1])


plt.tight_layout()
plt.show()
```



In [67]:
```python
first_product.describe()
```

Out[67]:

	Age	Education	Usage	Fitness	Income	Miles
count	80.000000	80.000000	80.000000	80.00000	80.00000	80.000000
mean	28.550000	15.037500	3.087500	2.96250	46418.02500	82.787500
std	7.221452	1.216383	0.782624	0.66454	9075.78319	28.874102
min	18.000000	12.000000	2.000000	1.00000	29562.00000	38.000000
25%	23.000000	14.000000	3.000000	3.00000	38658.00000	66.000000
50%	26.000000	16.000000	3.000000	3.00000	46617.00000	85.000000
75%	33.000000	16.000000	4.000000	3.00000	53439.00000	94.000000
max	50.000000	18.000000	5.000000	5.00000	68220.00000	188.000000

Comments

* It is clear from the abvoe graph that there are no outliers in the given data set, so we will move to the descriptive statistics

* We will start with the univariate analysis and once we determine the influencing factors we will switch to bi-variate analysis

* Under univariate analysis, we can address the following question:

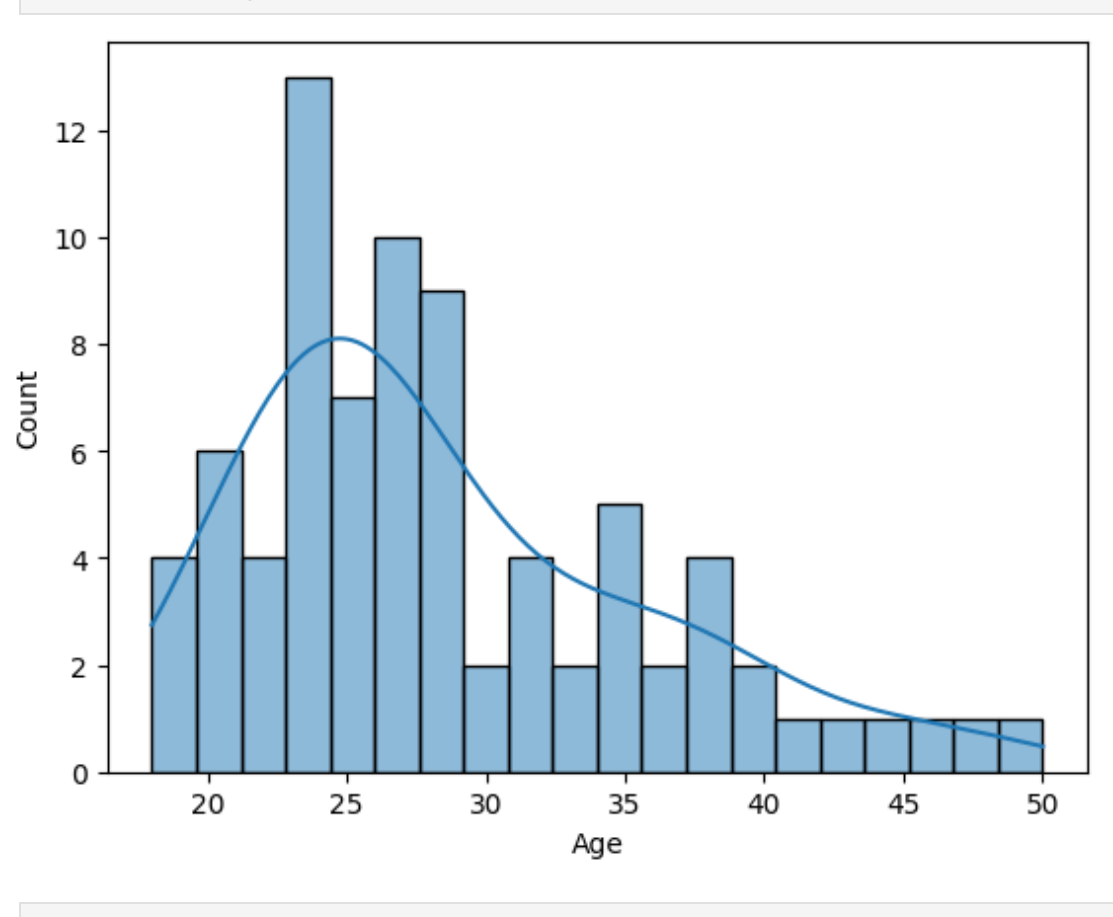
- Age: What is the typical age range of customers? Are there any age groups that buy more than others?
- Gender: What is the gender distribution of customers?
- Education: What are the common education levels of customers?
- MaritalStatus: What is the marital status distribution of customers? Are they mostly single or partnered?
- Usage: How often do customers plan to use the treadmill per week? What is the most common planned usage?
- Fitness: How do customers rate their own fitness level? What is the average self-rated fitness level?
- Income: What is the income distribution of customers? What is the average income?

In [ ]:
```python
first_product.head(5)
```

Out[ ]:

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	Age_Group
0	KP281	18	Male	14	Single	3	4	29562	112	NaN
1	KP281	19	Male	15	Single	2	3	31836	75	NaN
2	KP281	19	Female	14	Partnered	4	3	30699	66	NaN
3	KP281	19	Male	12	Single	3	3	32973	85	NaN
4	KP281	20	Male	13	Partnered	4	2	35247	47	NaN

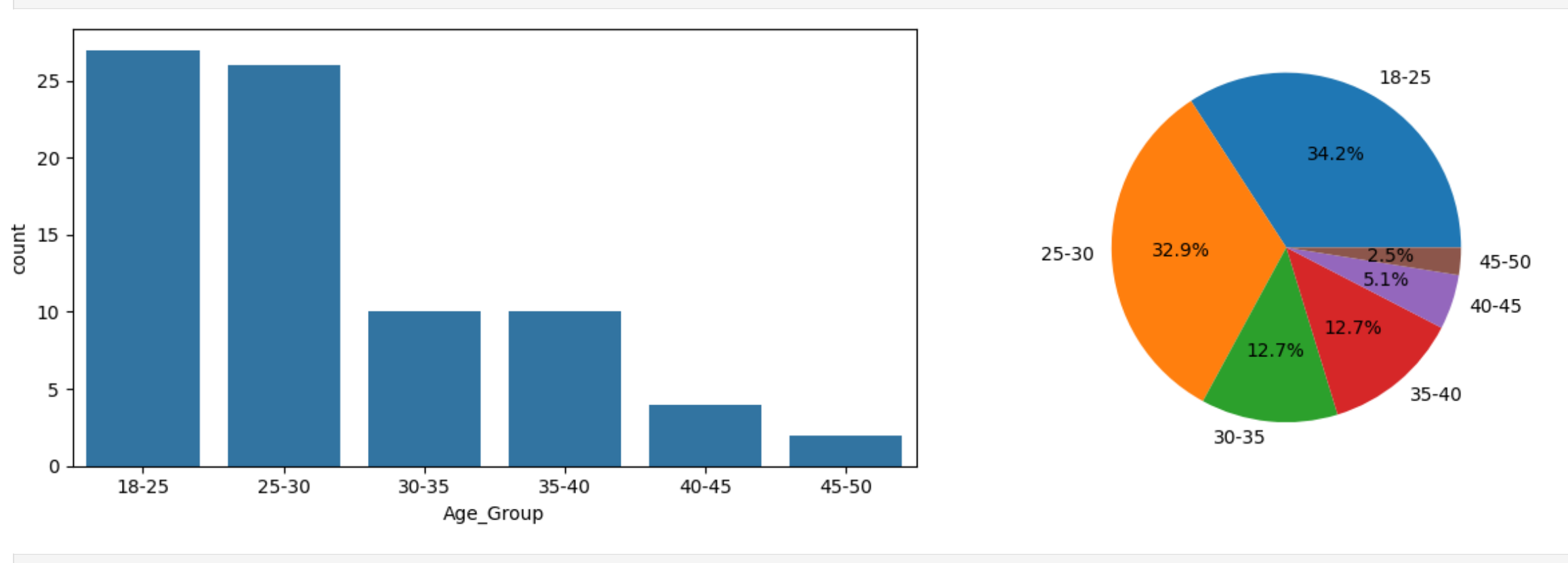
In [72]:
```python
# Age: What is the typical age range of customers? Are there any age groups that buy more than others?

sns.histplot(first_product['Age'],bins=20,kde=True)
plt.show()

# From the graph its clear that the product is famous amongst the age group between 25-30, however for better clarity lets
# divide the age into bins and understand their distribution
```



In [79]:
```python
# Creating bins for age group

bins = [18, 25, 30, 35, 40, 45,50]
labels = ['18-25', '25-30', '30-35', '35-40','40-45','45-50']

first_product['Age_Group'] = pd.cut(first_product['Age'], bins=bins, labels=labels, right=False)
```

```
/tmp/ipython-input-885679384.py:6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  first_product['Age_Group'] = pd.cut(first_product['Age'], bins=bins, labels=labels, right=False)
```

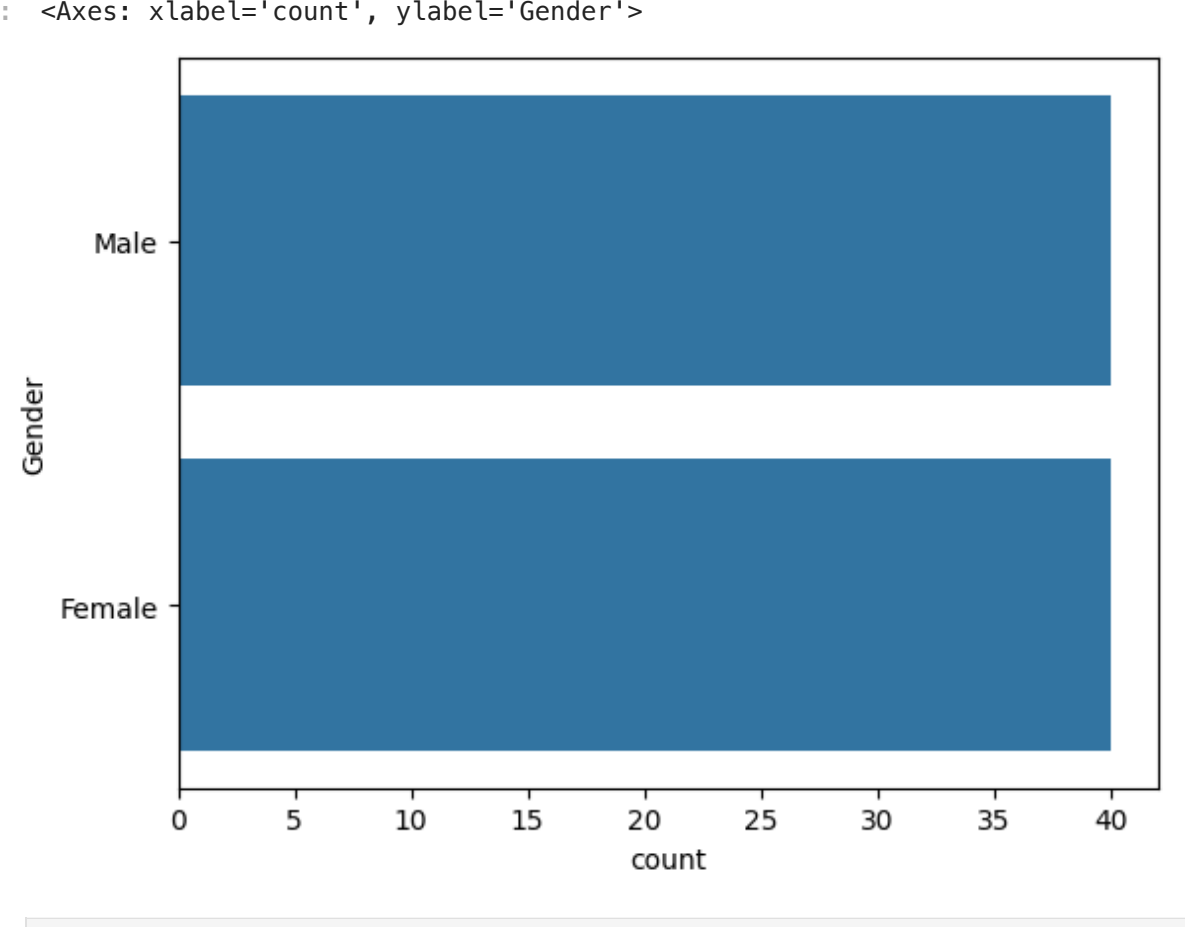
In [97]:
```python
# Lets understand the distribution of the data as per age group for better clarity

fig, ax = plt.subplots(1, 2, figsize=(13, 4))

sns.countplot(x='Age_Group',data=first_product, ax=ax[0])
ax[1].pie(first_product['Age_Group'].value_counts(),labels=first_product['Age_Group'].value_counts().index,autopct='%1.1f%%')

plt.tight_layout()
plt.show()
```



In [ ]:
```python
# It is clear from above graph that customers from age groups between 18-30 consitiues major portion (65%)
```

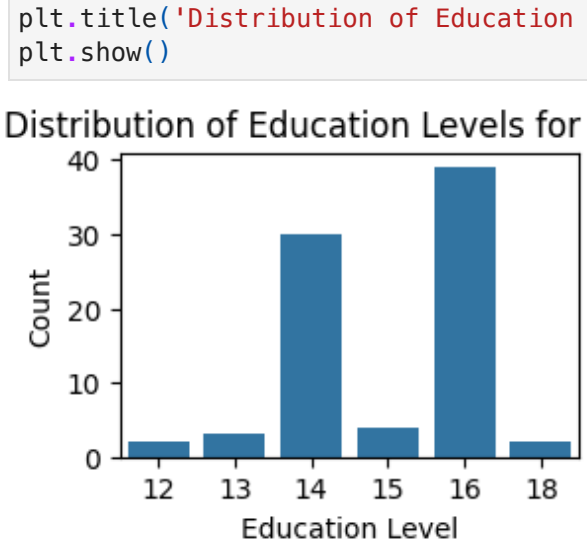
In [98]:
```python
# Gender: What is the gender distribution of customers?

sns.countplot(first_product['Gender'])
```

Out[98]:
```
<Axes: xlabel='count', ylabel='Gender'>
```



In [ ]:
```python
# There is an equal proportion between both genders, hence there is no clear winner
```

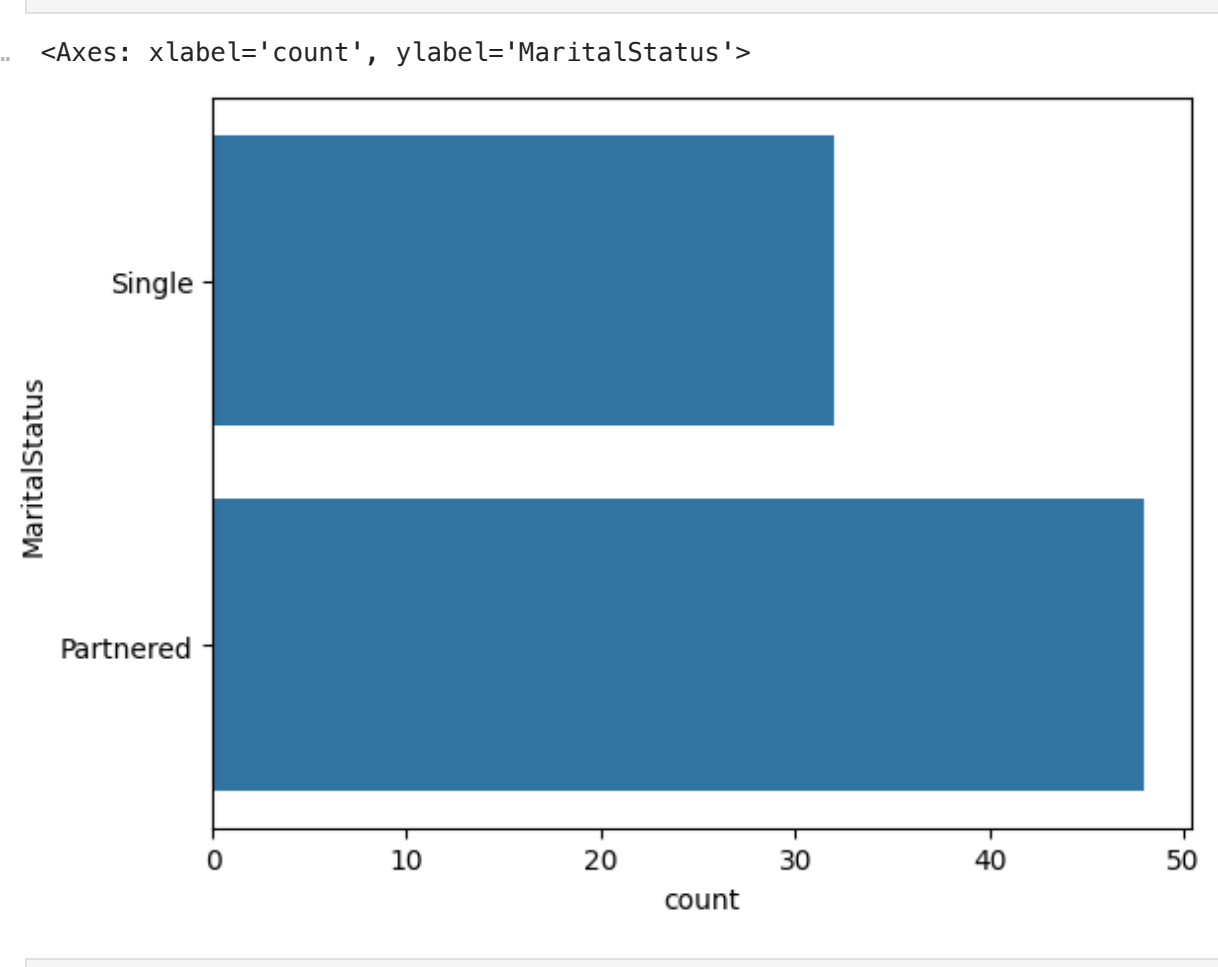
In [107…
```python
# Education: What are the common education levels of customers?

education_counts = first_product['Education'].value_counts().sort_index()  # Taking discrete values of the Education and then plotting it

plt.figure(figsize=(3, 2))
sns.barplot(x=education_counts.index, y=education_counts.values)
plt.xlabel('Education Level')
plt.ylabel('Count')
plt.title('Distribution of Education Levels for KP281')
plt.show()
```



In [113…
```python
# the above graphs shows that education grup 16 and 14 are most preferred ones
```

In [114…
```python
# MaritalStatus: What is the marital status distribution of customers? Are they mostly single or partnered?

sns.countplot(first_product['MaritalStatus'])
```

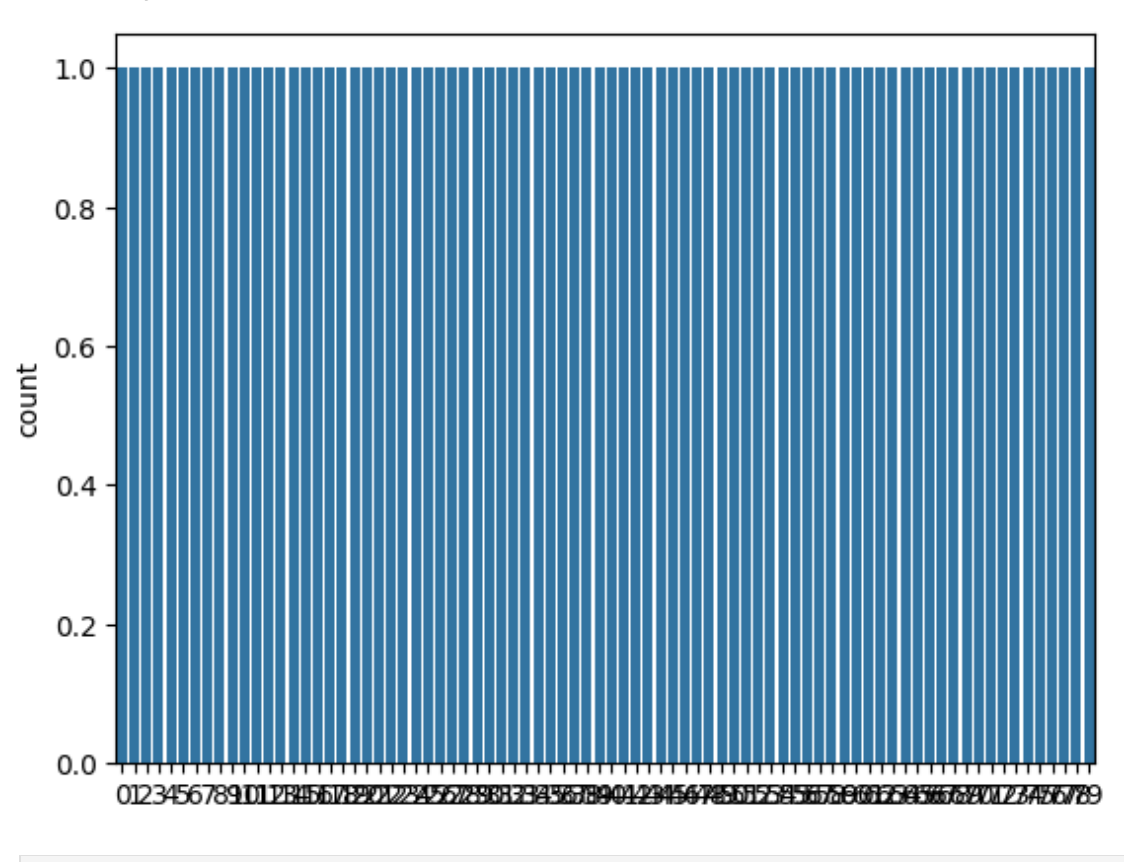
Out[114…
```
<Axes: xlabel='count', ylabel='MaritalStatus'>
```



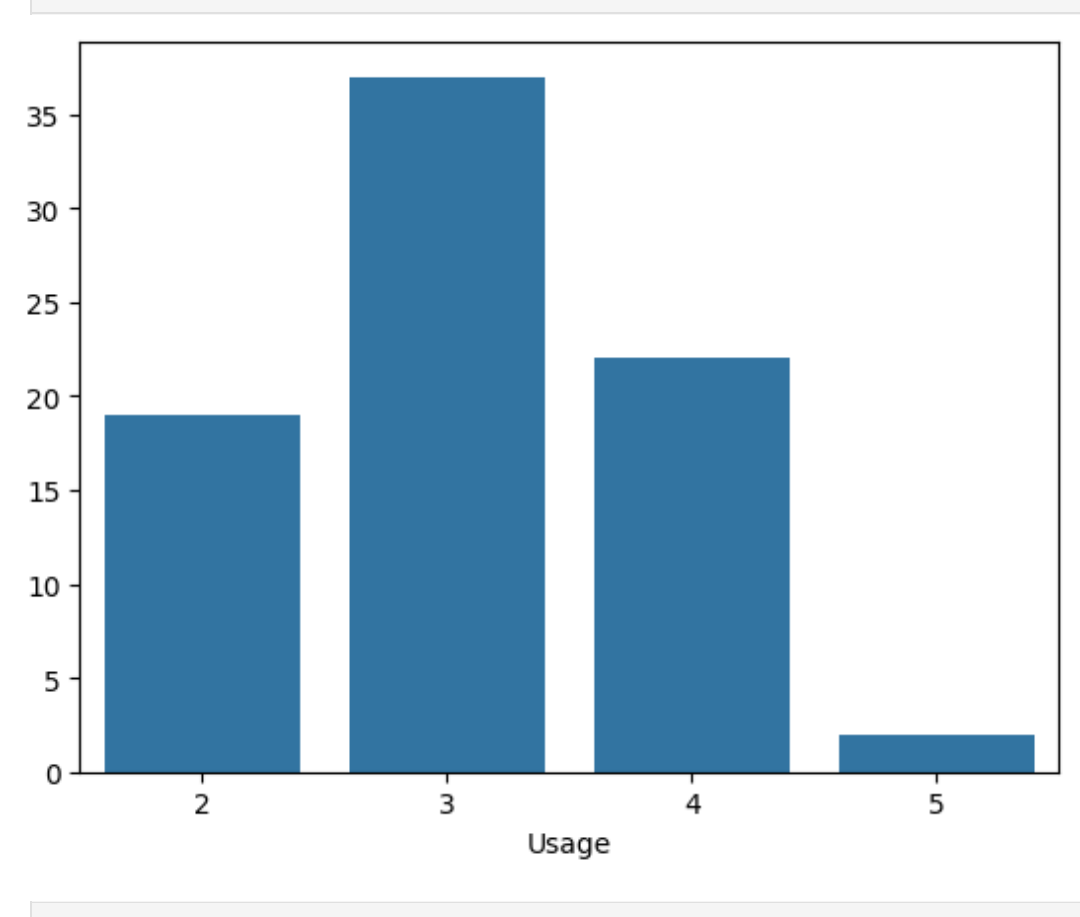
In [ ]:
```python
# Partenred i.e. Maried couples are frequent buyer of this product
```

In [116…
```python
# Usage: How often do customers plan to use the treadmill per week? What is the most common planned usage?

sns.countplot(first_product['Usage'])

# Countplot might be interpreting the numerical data as continous instead of discrete.
# This is causing it to group all values together. Lets calculate the data and use it
# to plot the graph
```

Out[116…
```
<Axes: ylabel='count'>
```



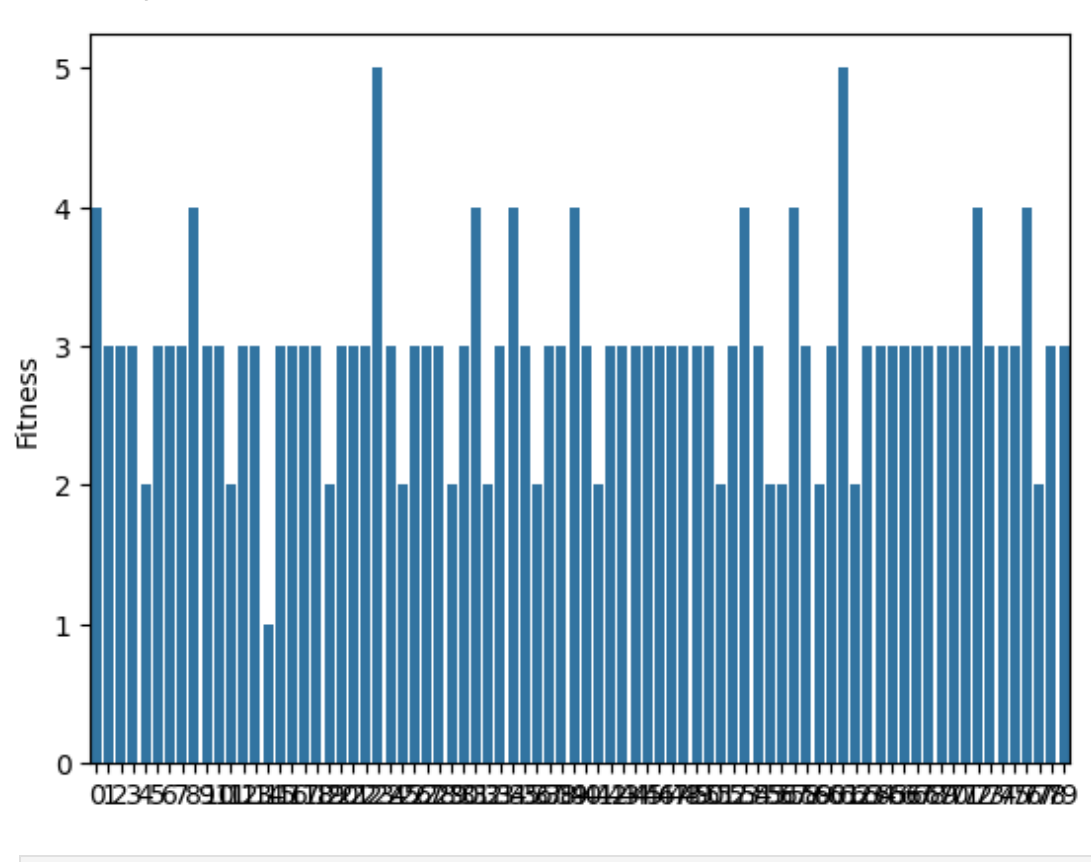
In [121…
```python
usage_counts = first_product['Usage'].value_counts().sort_index()
usage_counts

sns.barplot(x = usage_counts.index,y=usage_counts.values)
plt.show()
```



In [122…
```python
# It is famous amonst the people who wants to use it 3 times in a week
```

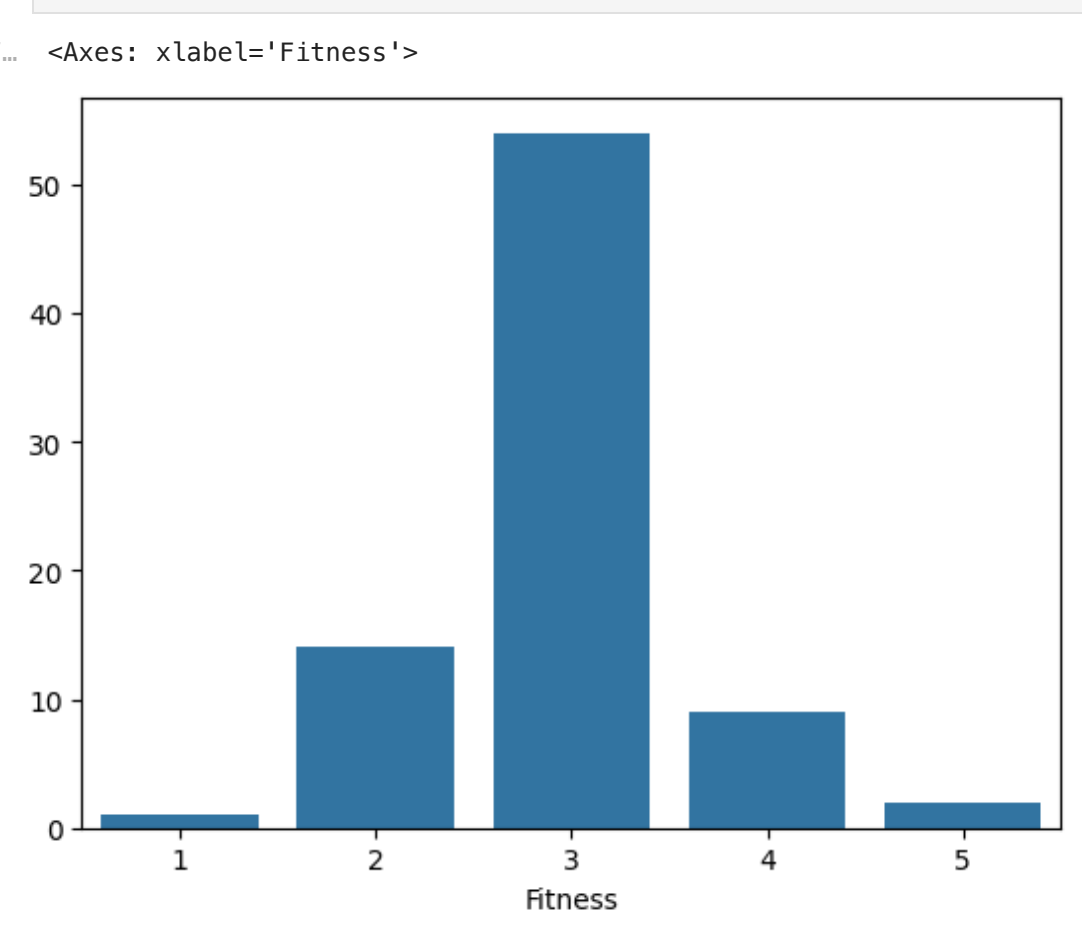
In [124…
```python
# Fitness: How do customers rate their own fitness level? What is the average self-rated fitness level?

sns.countplot(first_product['Fitness'])

# Repeatig the similar activity to convert continous to discrete
```

Out[124…
```
<Axes: ylabel='Fitness'>
```



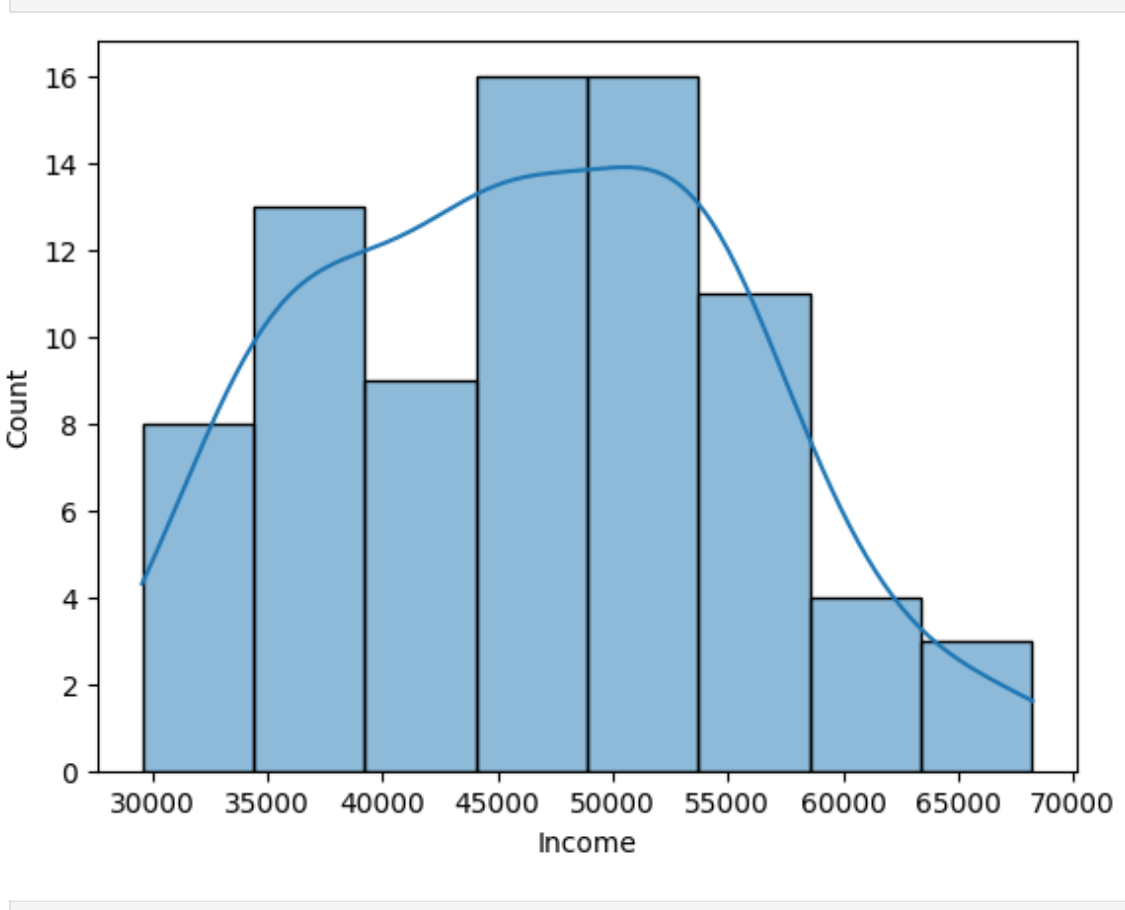
In [127…
```python
fitness_counts = first_product['Fitness'].value_counts().sort_index()


sns.barplot(x = fitness_counts.index,y=fitness_counts.values)
```

Out[127…
```
<Axes: xlabel='Fitness'>
```



In [128…
```python
# Customer who are rating themselves 3 are taking this product
```

In [130…
```python
# Income: What is the income distribution of customers? What is the average income?

sns.histplot(first_product['Income'],kde=True)
plt.show()
```



In [ ]:
```python
# Customers whose income is between $ 45000 - $55000 prefers the product
```

Summarizing the findings till now (Customer Profile for KP281):

Based on the univariate analysis for the KP281 product, the typical customer is between 18 and 30 years old, making up about 65% of buyers, with a nearly equal distribution between genders. They commonly have education levels of 14 or 16 years and are more likely to be partnered. These customers primarily plan to use the treadmill 3 times per week and rate their fitness level as 3. Their income typically falls between 45, 000and55,000.

In [ ]:

In [ ]:

In [ ]:

In [ ]:

KP481 Product

In [132…
```python
second_product = df[df['Product'] == 'KP481']
second_product.shape # All the data belong to KP281 are filtered
```

Out[132…
```
(60, 9)
```

In [133…
```python
second_product.head()
```

Out[133…

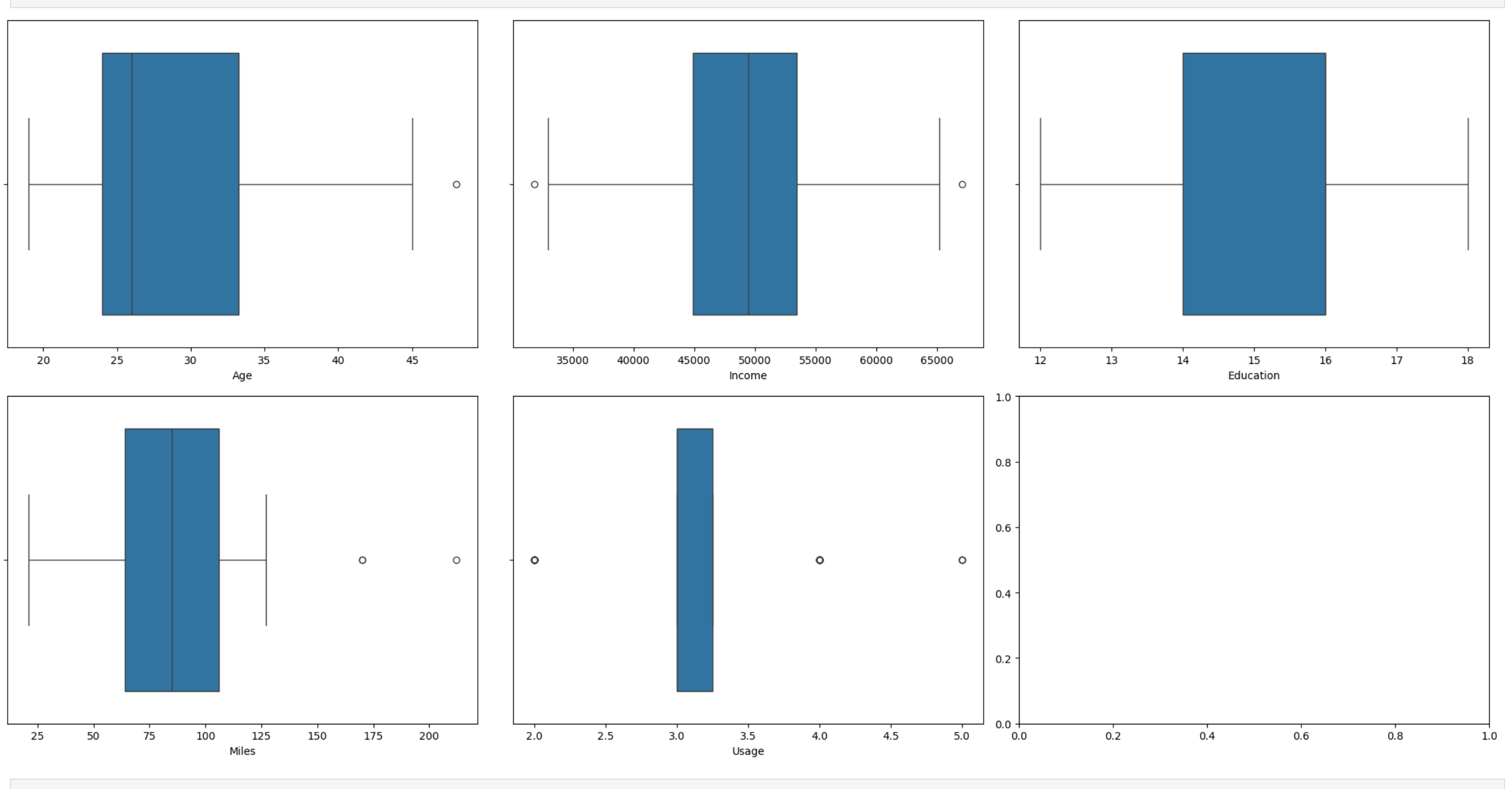
	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
80	KP481	19	Male	14	Single	3	3	31836	64
81	KP481	20	Male	14	Single	2	3	32973	53
82	KP481	20	Female	14	Partnered	3	3	34110	106
83	KP481	20	Male	14	Single	3	3	38658	95
84	KP481	21	Female	14	Partnered	5	4	34110	212

In [134…
```python
second_product.describe()
```

Out[134…

	Age	Education	Usage	Fitness	Income	Miles
count	60.000000	60.000000	60.000000	60.00000	60.000000	60.000000
mean	28.900000	15.116667	3.066667	2.90000	48973.650000	87.933333
std	6.645248	1.222552	0.799717	0.62977	8653.989388	33.263135
min	19.000000	12.000000	2.000000	1.00000	31836.000000	21.000000
25%	24.000000	14.000000	3.000000	3.00000	44911.500000	64.000000
50%	26.000000	16.000000	3.000000	3.00000	49459.500000	85.000000
75%	33.250000	16.000000	3.250000	3.00000	53439.000000	106.000000
max	48.000000	18.000000	5.000000	4.00000	67083.000000	212.000000

In [135…
```python
# First check for outliers for all columns

fig, ax = plt.subplots(2, 3, figsize=(20, 10))

sns.boxplot(x=second_product['Age'], ax=ax[0, 0])
sns.boxplot(x=second_product['Income'], ax=ax[0, 1])
sns.boxplot(x=second_product['Education'], ax=ax[0, 2])
sns.boxplot(x=second_product['Miles'], ax=ax[1, 0])
sns.boxplot(x=second_product['Usage'], ax=ax[1, 1])


plt.tight_layout()
plt.show()
```



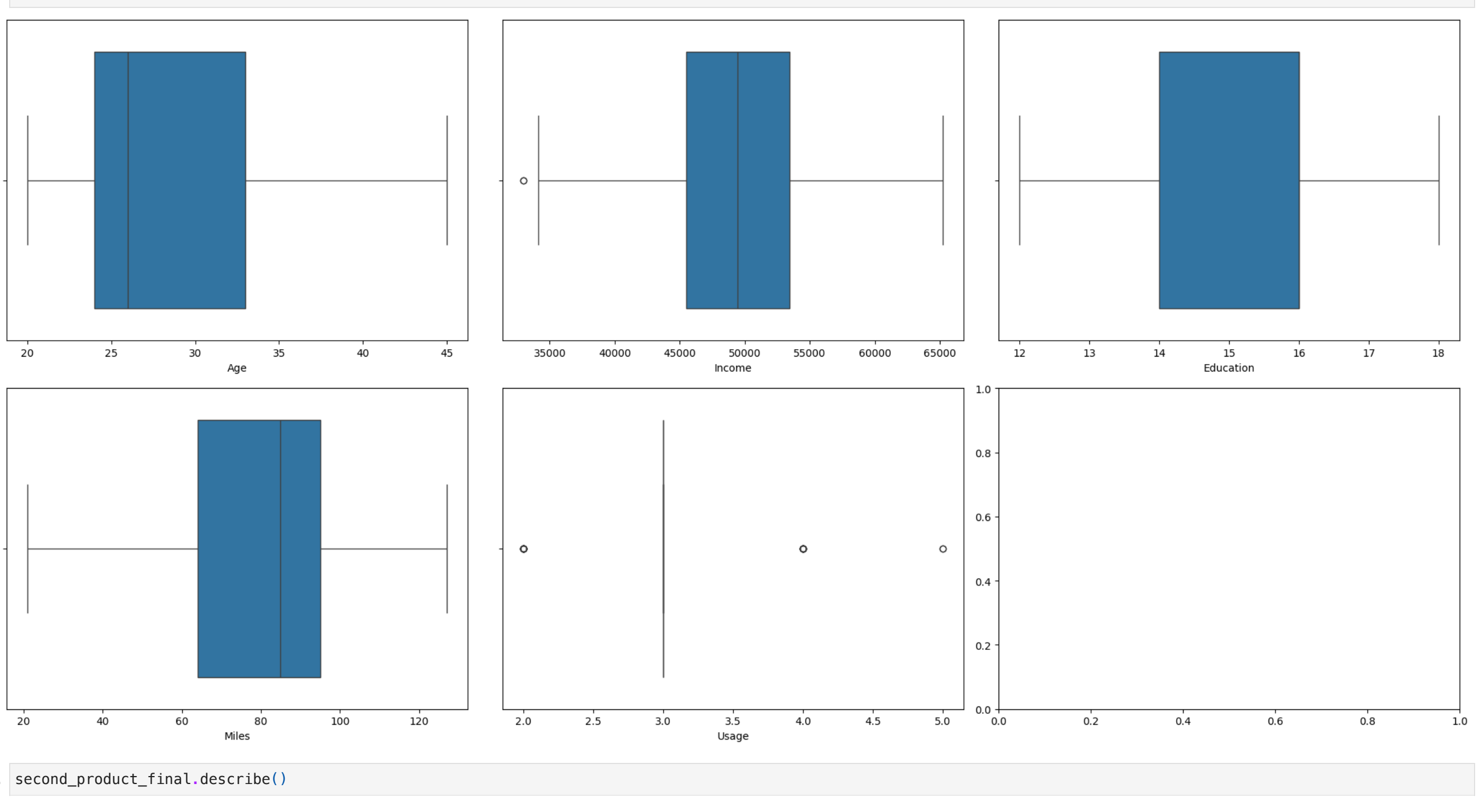
In [ ]:
```python
# There are some outliers so let's treat them
```

In [141…
```python
# Outliers treatment for Age

q1 = second_product['Age'].quantile(0.25)
q3 = second_product['Age'].quantile(0.75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

second_product_mod1 = second_product[(second_product['Age'] >= lower_bound) & (second_product['Age'] <= upper_bound)]
```

In [142…
```python
# Outliers treatment for Income

q1 = second_product['Income'].quantile(0.25)
q3 = second_product['Income'].quantile(0.75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

second_product_mod2 = second_product_mod1[(second_product_mod1['Income'] >= lower_bound) & (second_product_mod1['Income'] <= upper_bound)]
```

In [143…
```python
# Outliers treatment for Miles

q1 = second_product_mod2['Miles'].quantile(0.25)
q3 = second_product_mod2['Miles'].quantile(0.75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

second_product_final = second_product_mod2[(second_product_mod2['Miles'] >= lower_bound) & (second_product_mod2['Miles'] <= upper_bound)]
```

In [144…
```python
# Redrawing the box plot

fig, ax = plt.subplots(2, 3, figsize=(20, 10))

sns.boxplot(x=second_product_final['Age'], ax=ax[0, 0])
sns.boxplot(x=second_product_final['Income'], ax=ax[0, 1])
sns.boxplot(x=second_product_final['Education'], ax=ax[0, 2])
sns.boxplot(x=second_product_final['Miles'], ax=ax[1, 0])
sns.boxplot(x=second_product_final['Usage'], ax=ax[1, 1])


plt.tight_layout()
plt.show()
```



In [145…
```python
second_product_final.describe()
```

Out[145…

	Age	Education	Usage	Fitness	Income	Miles
count	54.000000	54.000000	54.000000	54.000000	54.000000	54.000000
mean	28.777778	15.203704	3.018519	2.851852	49046.000000	83.537037
std	6.175627	1.219026	0.764563	0.626684	8082.541899	25.409635
min	20.000000	12.000000	2.000000	1.000000	32973.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	45480.000000	64.000000
50%	26.000000	16.000000	3.000000	3.000000	49459.500000	85.000000
75%	33.000000	16.000000	3.000000	3.000000	53439.000000	95.000000
max	45.000000	18.000000	5.000000	4.000000	65220.000000	127.000000

Comments

Post Outliers are treated, lets dig into uni-variate analysis

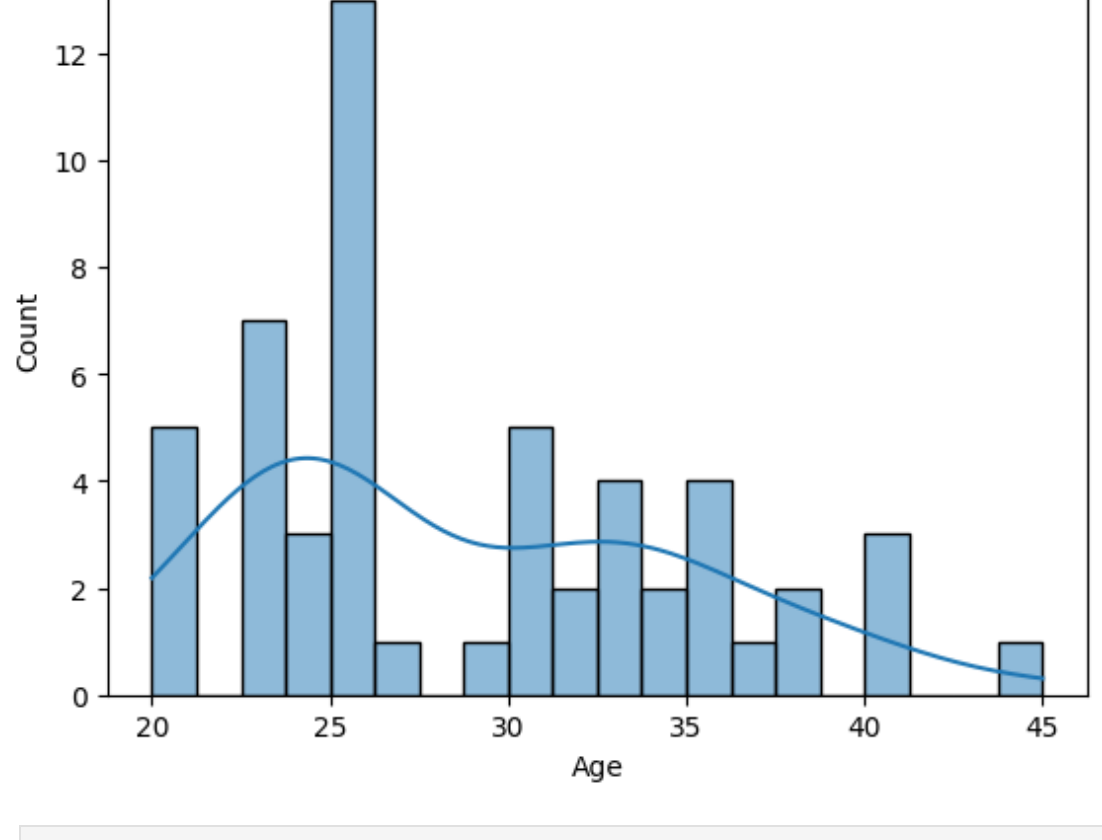
- Age: What is the typical age range of customers? Are there any age groups that buy more than others?
- Gender: What is the gender distribution of customers?
- Education: What are the common education levels of customers?
- MaritalStatus: What is the marital status distribution of customers? Are they mostly single or partnered?
- Usage: How often do customers plan to use the treadmill per week? What is the most common planned usage?
- Fitness: How do customers rate their own fitness level? What is the average self-rated fitness level?
- Income: What is the income distribution of customers? What is the average income?

In [146…
```python
second_product_final.head(5)
```

Out[146…

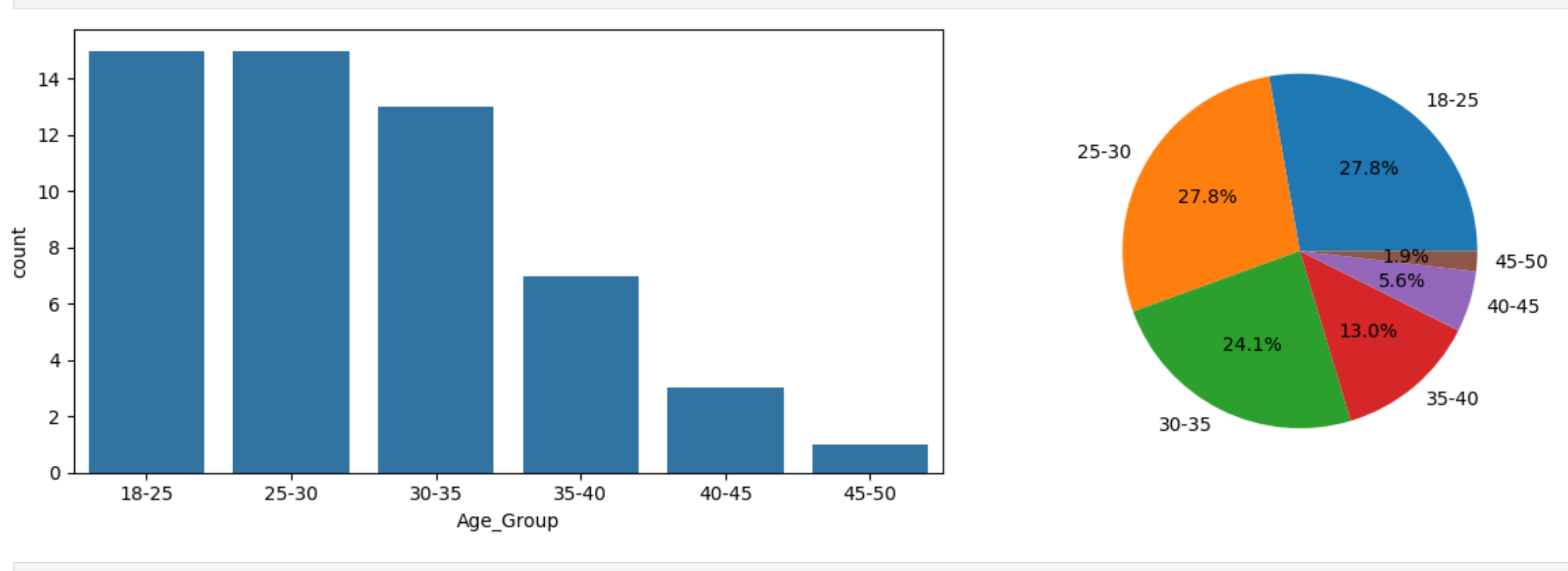
	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
81	KP481	20	Male	14	Single	2	3	32973	53
82	KP481	20	Female	14	Partnered	3	3	34110	106
83	KP481	20	Male	14	Single	3	3	38658	95
85	KP481	21	Male	16	Partnered	2	2	34110	42
86	KP481	21	Male	12	Partnered	2	2	32973	53

In [147…
```python
# Age: What is the typical age range of customers? Are there any age groups that buy more than others?

sns.histplot(second_product_final['Age'],bins=20,kde=True)
plt.show()

# From the graph its clear that the product is famous amongst the age group between 22-26, however for better clarity lets
# divide the age into bins and understand their distribution
```



In [148…
```
# Creating bins for age group

bins = [18, 25, 30, 35, 40, 45,50]
labels = ['18-25', '25-30', '30-35', '35-40','40-45','45-50']

second_product_final['Age_Group'] = pd.cut(second_product_final['Age'], bins=bins, labels=labels, right=False)
```

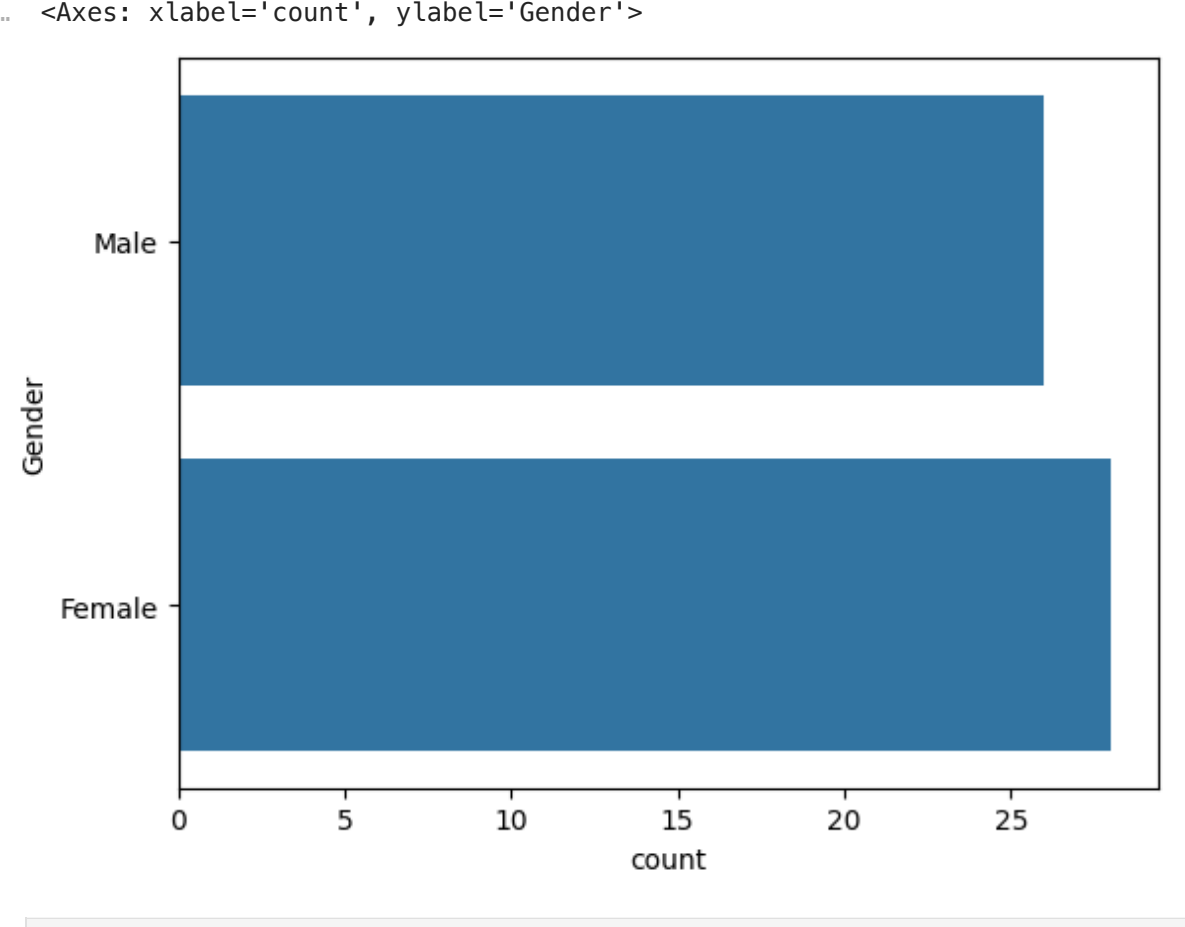
In [149…
```
# Lets understand the distribution of the data as per age group for better clarity

fig, ax = plt.subplots(1, 2, figsize=(13, 4))

sns.countplot(x='Age_Group',data=second_product_final, ax=ax[0])
ax[1].pie(second_product_final['Age_Group'].value_counts(),labels=second_product_final['Age_Group'].value_counts().index,autopct='%1.1f%%')

plt.tight_layout()
plt.show()
```



In [ ]:
```
# It is clear from above graph that customers from age groups between 18-35 consitiutes major portion (78%)
```

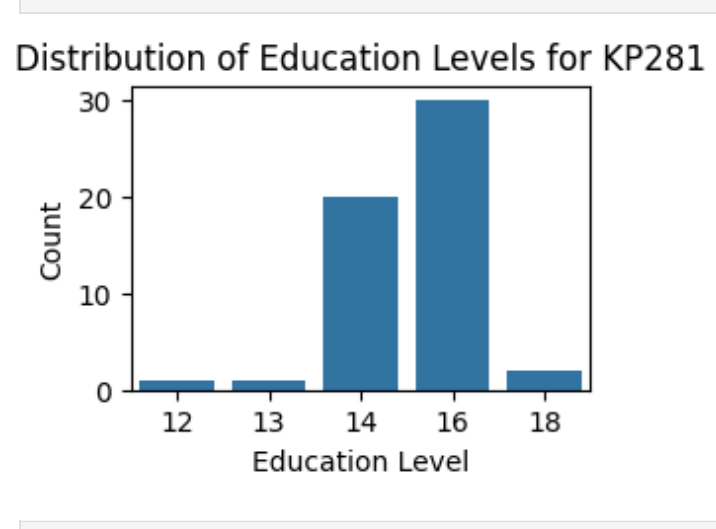
In [150…
```
# Gender: What is the gender distribution of customers?

sns.countplot(second_product_final['Gender'])
```

Out[150…
```
<Axes: xlabel='count', ylabel='Gender'>
```



In [ ]:
```
# Female customers prefer this product more than their counterpart
```

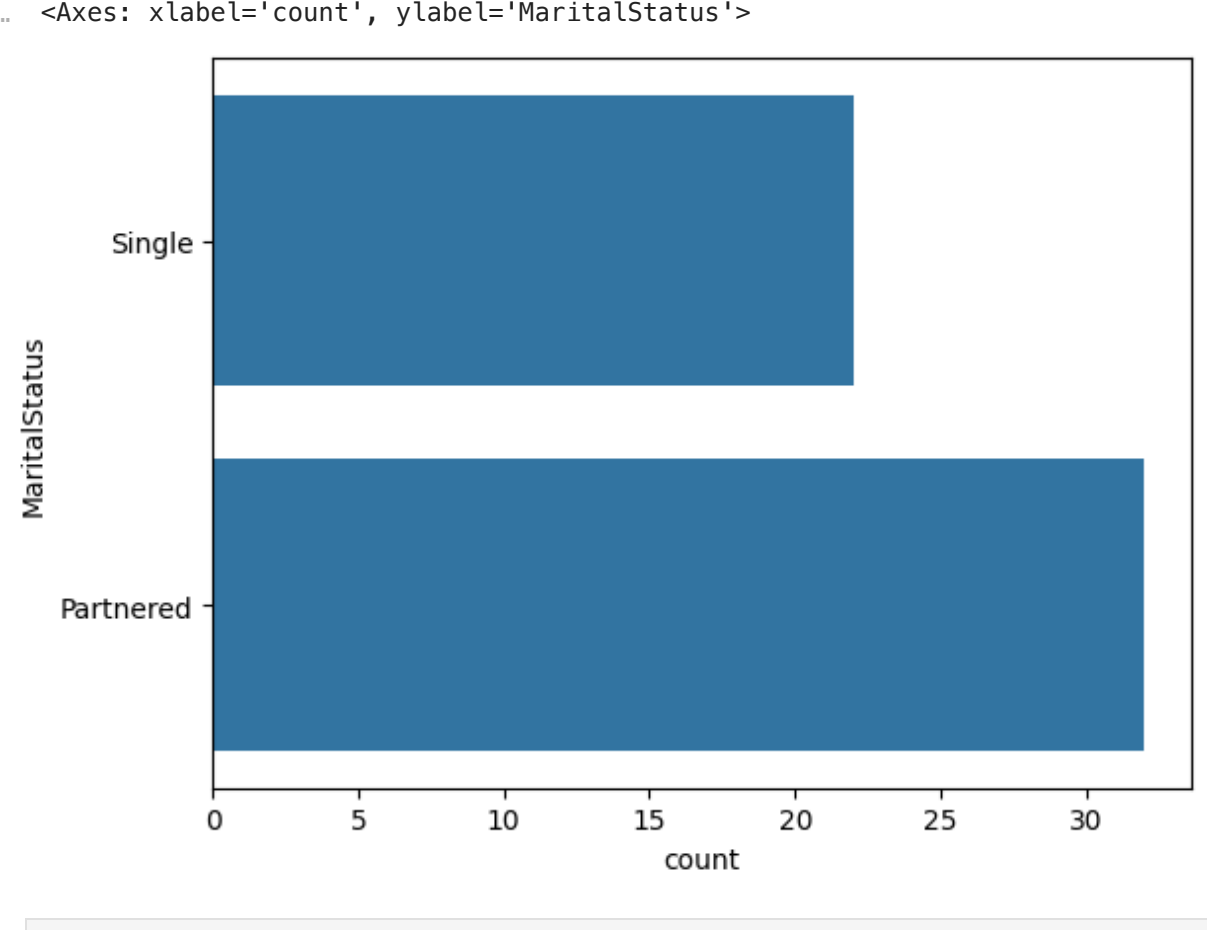
In [151…
```
# Education: What are the common education levels of customers?

education_counts = second_product_final['Education'].value_counts().sort_index()  # Taking discrete values of the Education and then plotting it

plt.figure(figsize=(3, 2))
sns.barplot(x=education_counts.index, y=education_counts.values)
plt.xlabel('Education Level')
plt.ylabel('Count')
plt.title('Distribution of Education Levels for KP281')
plt.show()
```


Distribution of Education Levels for KP281

In [ ]:
```
# the above graphs shows that education grup 16 and 14 are most preferred ones
```

In [152…
```
# MaritalStatus: What is the marital status distribution of customers? Are they mostly single or partnered?

sns.countplot(second_product_final['MaritalStatus'])
```

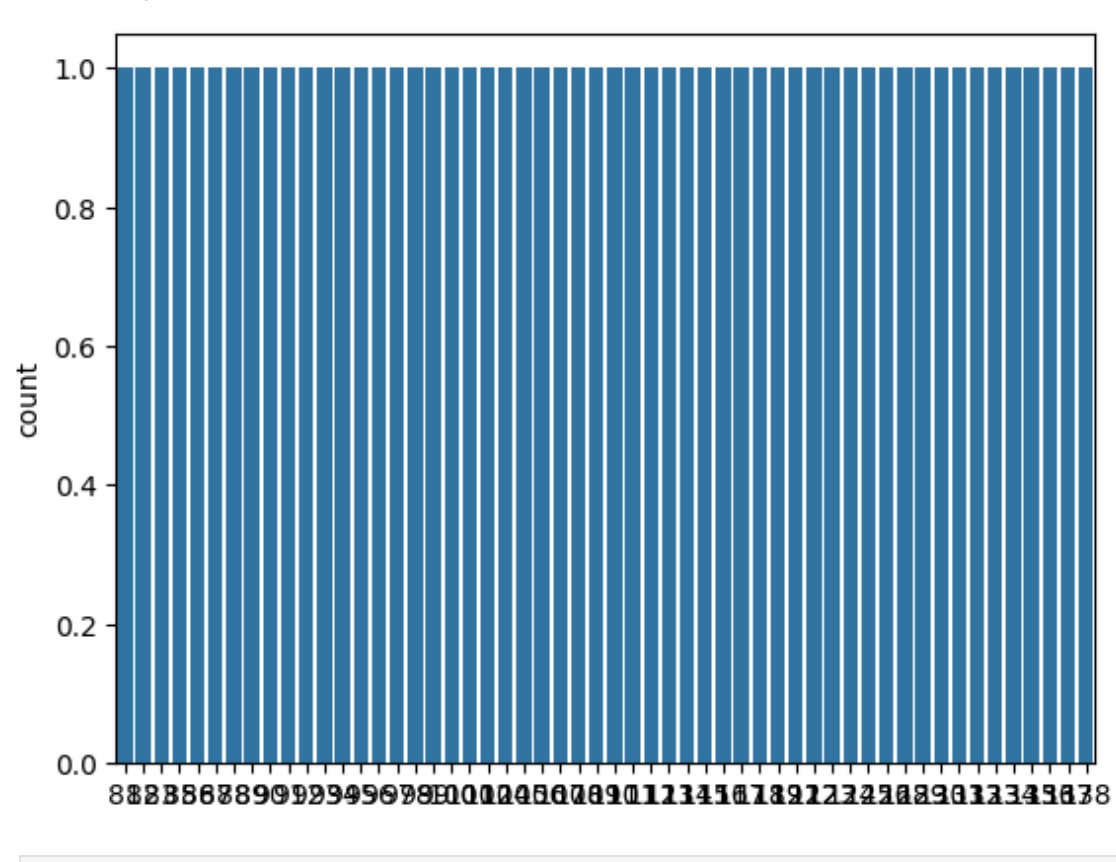
Out[152…
```
<Axes: xlabel='count', ylabel='MaritalStatus'>
```



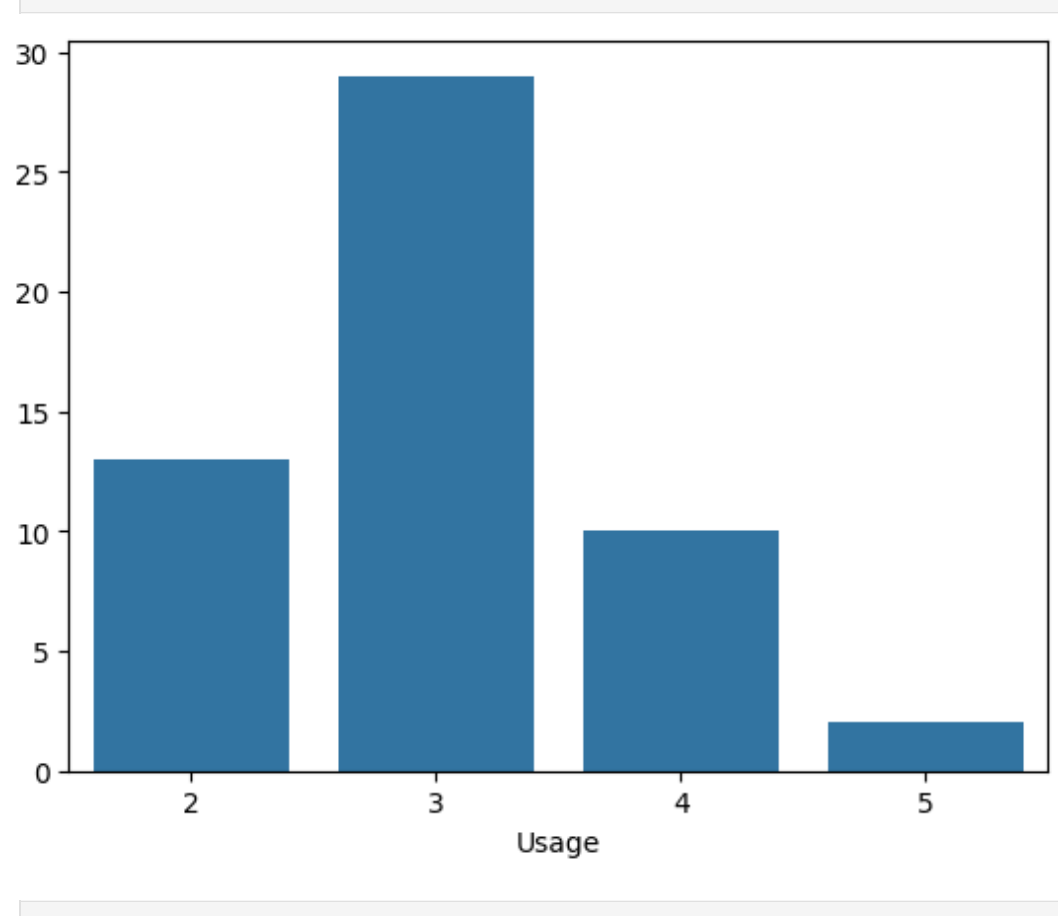
In [ ]:
```
# Partenred i.e. Maried couples are frequent buyer of this product
```

In [153…
```
# Usage: How often do customers plan to use the treadmill per week? What is the most common planned usage?

sns.countplot(second_product_final['Usage'])

# Countplot might be interpreting the numerical data as continous instead of discrete.
# This is causing it to group all values together. Lets calculate the data and use it
# to plot the graph
```

Out[153…
```
<Axes: ylabel='count'>
```



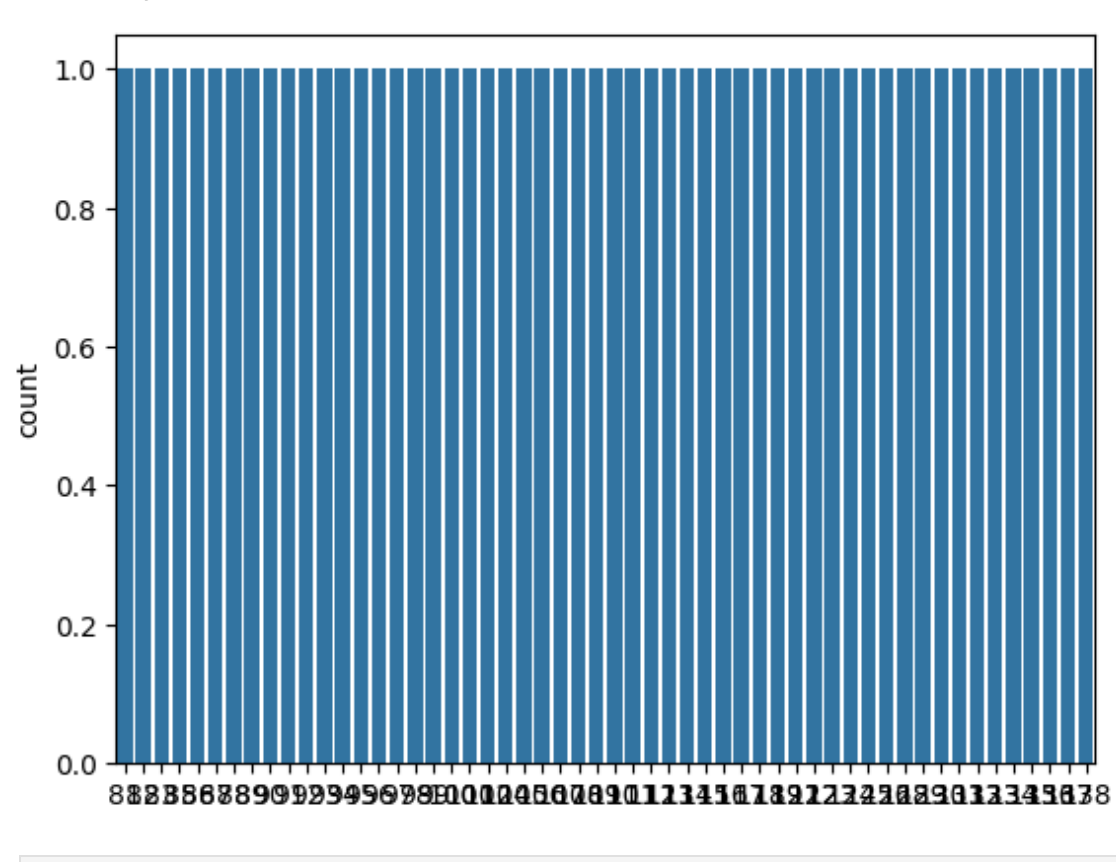
In [154…
```
usage_counts = second_product_final['Usage'].value_counts().sort_index()
usage_counts

sns.barplot(x = usage_counts.index,y=usage_counts.values)
plt.show()
```



In [ ]:
```
# It is famous amonst the people who wants to use it 3 times in a week
```

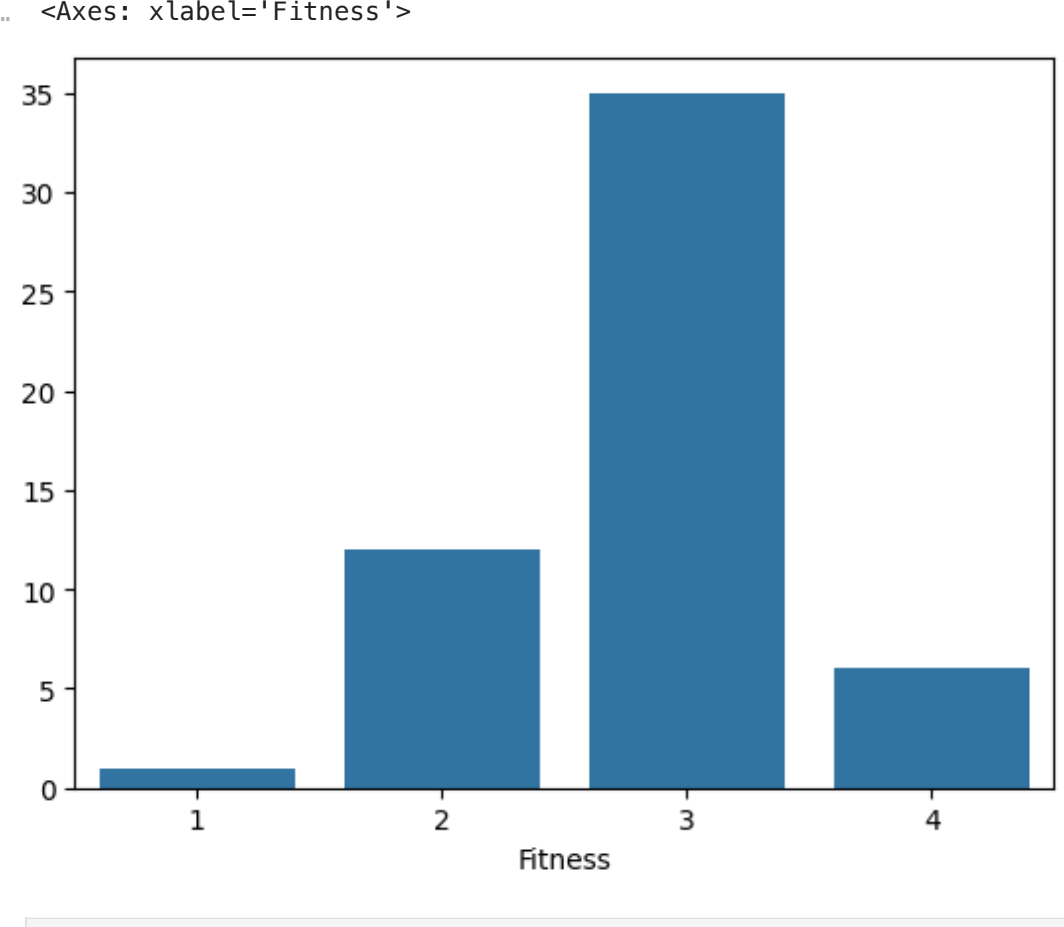
In [155…
```
# Fitness: How do customers rate their own fitness level? What is the average self-rated fitness level?

sns.countplot(second_product_final['Fitness'])

# Repeatig the similar activity to convert continous to discrete
```

Out[155…
```
<Axes: ylabel='count'>
```



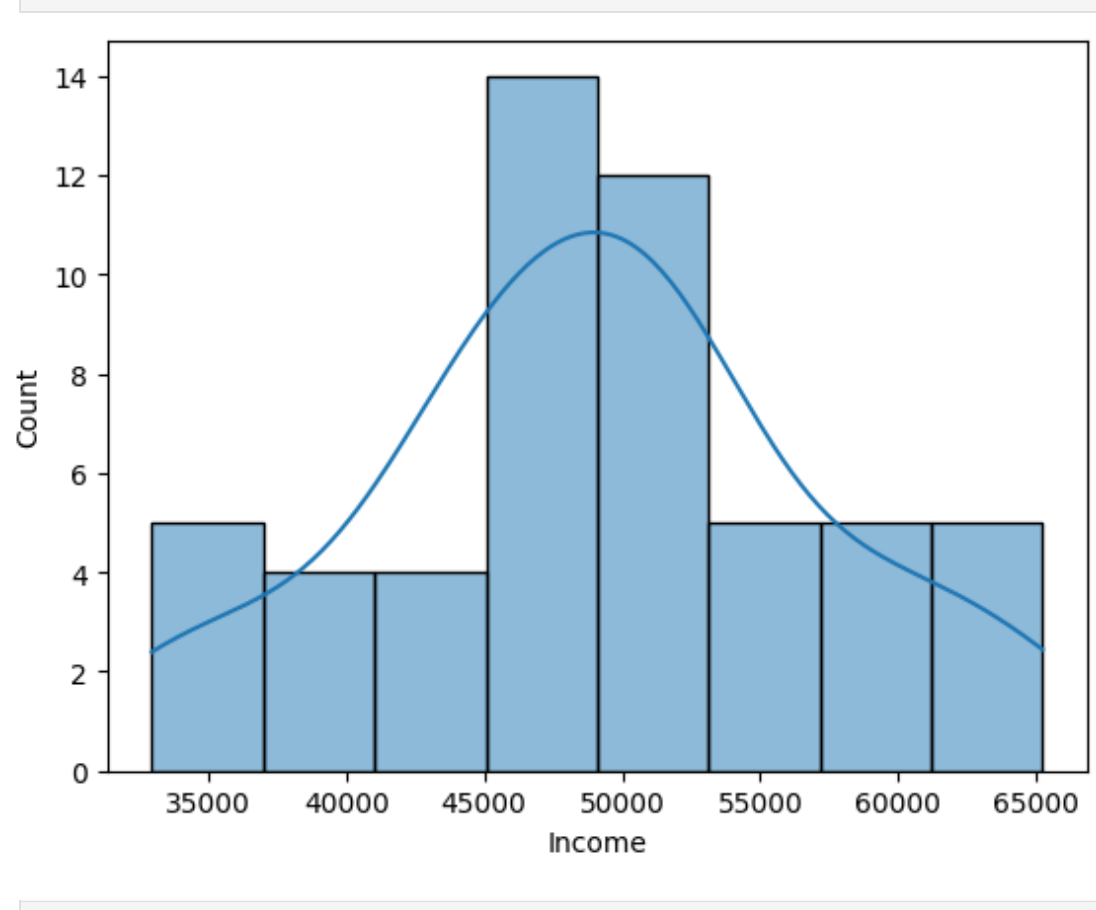
In [156…
```
fitness_counts = second_product_final['Fitness'].value_counts().sort_index()


sns.barplot(x = fitness_counts.index,y=fitness_counts.values)
```

Out[156…
```
<Axes: xlabel='Fitness'>
```



In [ ]:
```
# Customer who are rating themselves 3 are taking this product
```

In [157…
```
# Income: What is the income distribution of customers? What is the average income?

sns.histplot(second_product_final['Income'],kde=True)
plt.show()
```



In [ ]:
```
# Customers whose income is between $ 45000 - $55000 prefers the product
```

Summarizing the findings till now (Customer Profile for KP481):

Based on the univariate analysis for the KP481 product, the typical customer is between 18 and 35 years old, making up about 78% of buyers, with a higher preference among female customers. They are more likely to be partnered and primarily plan to use the treadmill 3 times per week. These customers typically rate their fitness level as 3 and have incomes ranging from $45,000to$55,000.

In [ ]:

In [ ]:

In [ ]:

In [ ]:

KP481 Product

In [158…
```
third_product = df[df['Product'] == 'KP781']
third_product.shape # All the data belong to KP281 are filtered
```

Out[158…
```
(40, 9)
```

In [159…
```
third_product.head()
```

Out[159…

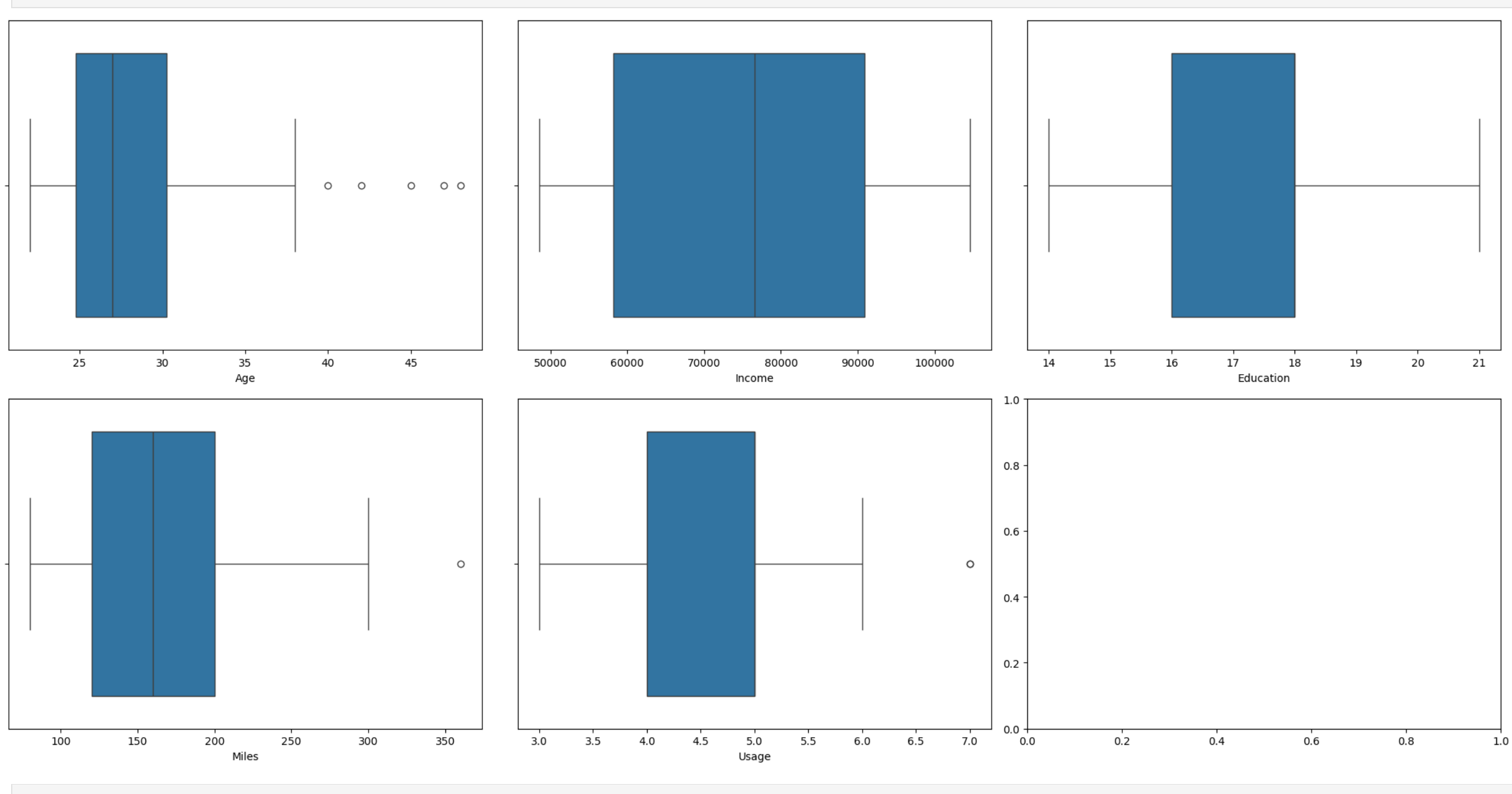
	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
140	KP781	22	Male	14	Single	4	3	48658	106
141	KP781	22	Male	16	Single	3	5	54781	120
142	KP781	22	Male	18	Single	4	5	48556	200
143	KP781	23	Male	16	Single	4	5	58516	140
144	KP781	23	Female	18	Single	5	4	53536	100

In [160…
```
third_product.describe()
```

Out[160…

	Age	Education	Usage	Fitness	Income	Miles
count	40.000000	40.000000	40.000000	40.000000	40.00000	40.000000
mean	29.100000	17.325000	4.775000	4.625000	75441.57500	166.900000
std	6.971738	1.639066	0.946993	0.667467	18505.83672	60.066544
min	22.000000	14.000000	3.000000	3.000000	48556.00000	80.000000
25%	24.750000	16.000000	4.000000	4.000000	58204.75000	120.000000
50%	27.000000	18.000000	5.000000	5.000000	76568.50000	160.000000
75%	30.250000	18.000000	5.000000	5.000000	90886.00000	200.000000
max	48.000000	21.000000	7.000000	5.000000	104581.00000	360.000000

In [161…
```
# First check for outliers for all columns

fig, ax = plt.subplots(2, 3, figsize=(20, 10))

sns.boxplot(x=third_product['Age'], ax=ax[0, 0])
sns.boxplot(x=third_product['Income'], ax=ax[0, 1])
sns.boxplot(x=third_product['Education'], ax=ax[0, 2])
sns.boxplot(x=third_product['Miles'], ax=ax[1, 0])
sns.boxplot(x=third_product['Usage'], ax=ax[1, 1])


plt.tight_layout()
plt.show()
```

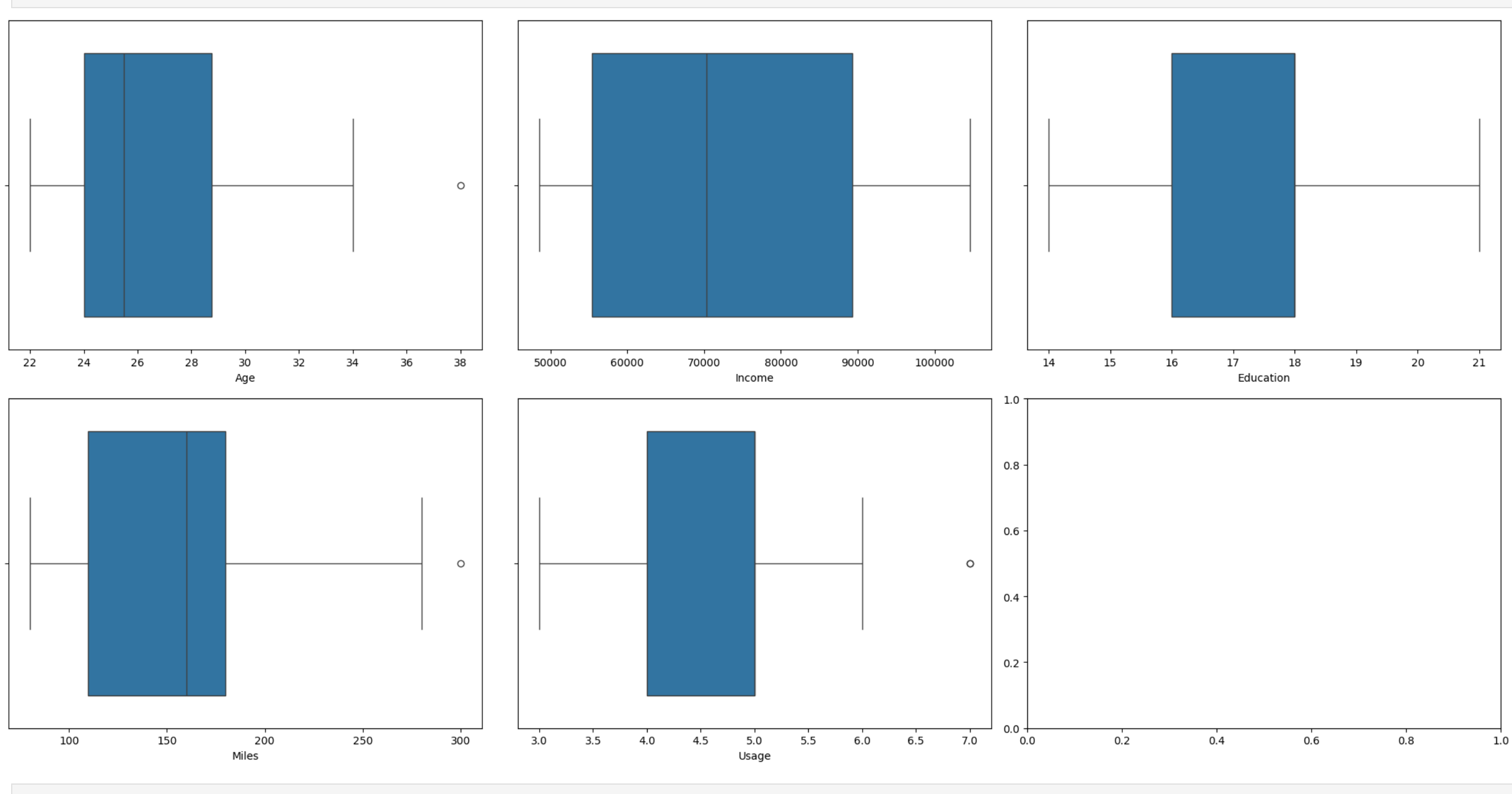


In [ ]:
```
# There are some outliers so let's treat them
```

In [162…
```
# Outliers treatment for Age

q1 = third_product['Age'].quantile(0.25)
q3 = third_product['Age'].quantile(0.75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

third_product_mod1 = third_product[(third_product['Age'] >= lower_bound) & (third_product['Age'] <= upper_bound)]
```

In [165…
```
# Outliers treatment for Miles

q1 = third_product_mod1['Miles'].quantile(0.25)
q3 = third_product_mod1['Miles'].quantile(0.75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

third_product_mod2 = third_product_mod1[(third_product_mod1['Miles'] >= lower_bound) & (third_product_mod1['Miles'] <= upper_bound)]
```

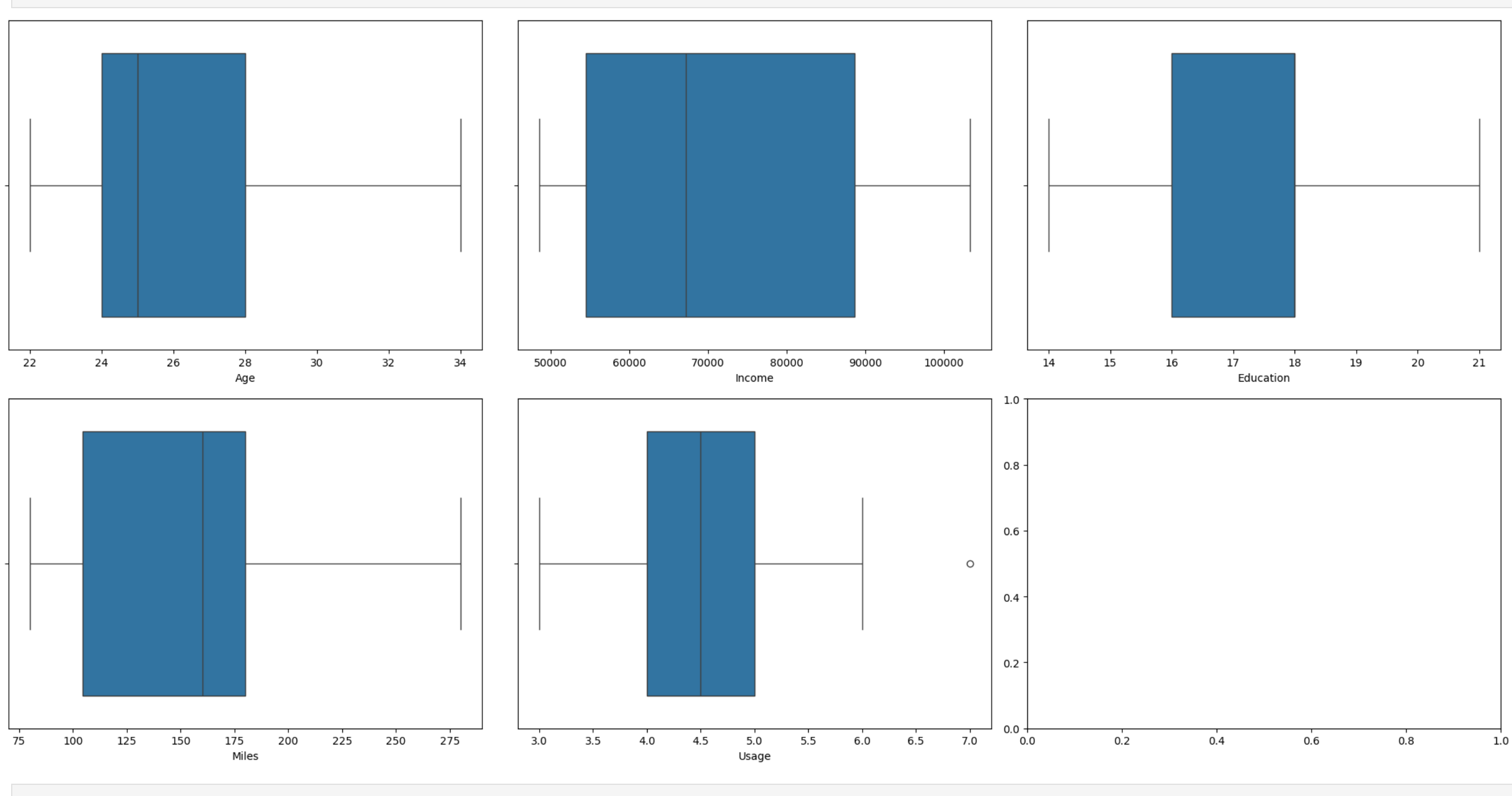
In [166…
```
# Redrawing the box plot

fig, ax = plt.subplots(2, 3, figsize=(20, 10))

sns.boxplot(x=third_product_mod2['Age'], ax=ax[0, 0])
sns.boxplot(x=third_product_mod2['Income'], ax=ax[0, 1])
sns.boxplot(x=third_product_mod2['Education'], ax=ax[0, 2])
sns.boxplot(x=third_product_mod2['Miles'], ax=ax[1, 0])
sns.boxplot(x=third_product_mod2['Usage'], ax=ax[1, 1])


plt.tight_layout()
plt.show()
```



In [170…
```
# Still there is an outlier for Age, lets treat it:

q1 = third_product_mod2['Age'].quantile(0.25)
q3 = third_product_mod2['Age'].quantile(0.75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

third_product_mod3 = third_product_mod2[(third_product_mod2['Age'] >= lower_bound) & (third_product_mod2['Age'] <= upper_bound)]
```

In [171…
```
# Still there is an outlier for Miles, lets treat it:

q1 = third_product_mod3['Miles'].quantile(0.25)
q3 = third_product_mod3['Miles'].quantile(0.75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

third_product_final = third_product_mod3[(third_product_mod3['Miles'] >= lower_bound) & (third_product_mod3['Miles'] <= upper_bound)]
```

In [172…
```
# Redrawing the box plot

fig, ax = plt.subplots(2, 3, figsize=(20, 10))

sns.boxplot(x=third_product_final['Age'], ax=ax[0, 0])
sns.boxplot(x=third_product_final['Income'], ax=ax[0, 1])
sns.boxplot(x=third_product_final['Education'], ax=ax[0, 2])
sns.boxplot(x=third_product_final['Miles'], ax=ax[1, 0])
sns.boxplot(x=third_product_final['Usage'], ax=ax[1, 1])


plt.tight_layout()
plt.show()
```



In [ ]:

In [ ]:

In [173…
```
third_product_final.describe()
```

Out[173…

	Age	Education	Usage	Fitness	Income	Miles
count	32.00000	32.000000	32.00000	32.000000	32.000000	32.000000
mean	26.25000	17.312500	4.71875	4.562500	70969.156250	156.437500
std	3.14181	1.554131	0.92403	0.715609	17710.424027	49.938632
min	22.00000	14.000000	3.00000	3.000000	48556.000000	80.000000
25%	24.00000	16.000000	4.00000	4.000000	54469.750000	104.500000
50%	25.00000	18.000000	4.50000	5.000000	67231.000000	160.000000
75%	28.00000	18.000000	5.00000	5.000000	88707.250000	180.000000
max	34.00000	21.000000	7.00000	5.000000	103336.000000	280.000000

Comments

Post Outliers are treated, lets dig into uni-variate analysis

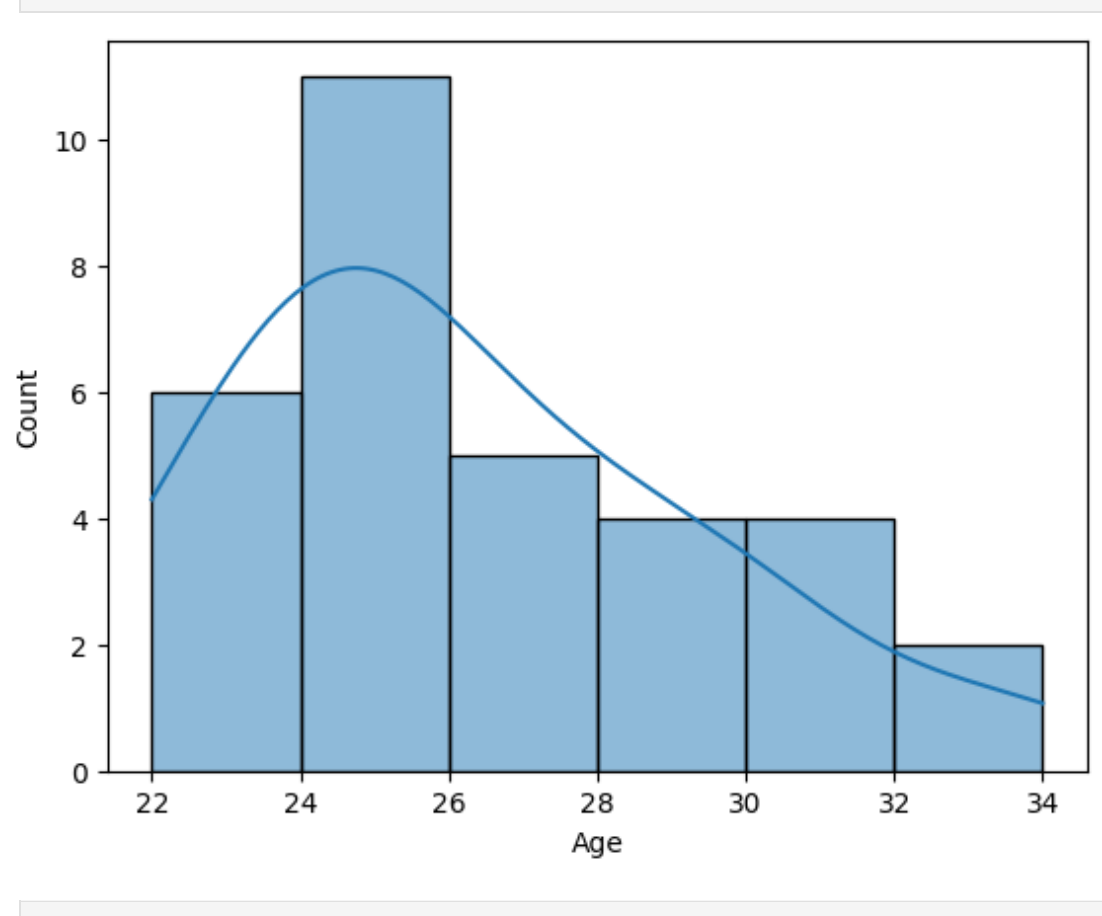
- Age: What is the typical age range of customers? Are there any age groups that buy more than others?
- Gender: What is the gender distribution of customers?
- Education: What are the common education levels of customers?
- MaritalStatus: What is the marital status distribution of customers? Are they mostly single or partnered?
- Usage: How often do customers plan to use the treadmill per week? What is the most common planned usage?
- Fitness: How do customers rate their own fitness level? What is the average self-rated fitness level?
- Income: What is the income distribution of customers? What is the average income?

In [174…
```
third_product_final.head(5)
```

Out[174…

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
140	KP781	22	Male	14	Single	4	3	48658	106
141	KP781	22	Male	16	Single	3	5	54781	120
142	KP781	22	Male	18	Single	4	5	48556	200
143	KP781	23	Male	16	Single	4	5	58516	140
144	KP781	23	Female	18	Single	5	4	53536	100

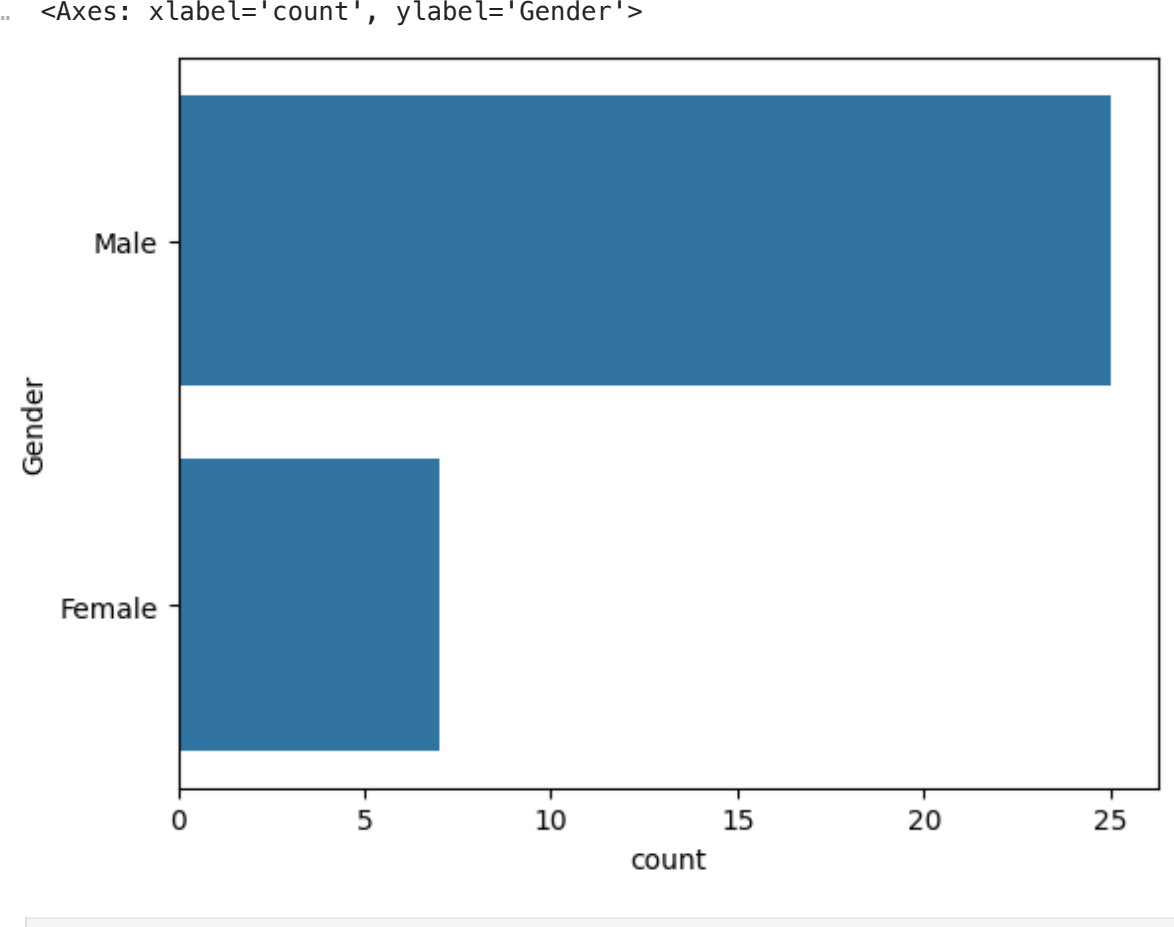
In [176…
```
# Age: What is the typical age range of customers? Are there any age groups that buy more than others?

sns.histplot(third_product_final['Age'],kde=True)
plt.show()
```



In [ ]:
```
# It is clear from above graph that customers from age groups between 24-26 consitiutes major portion
```

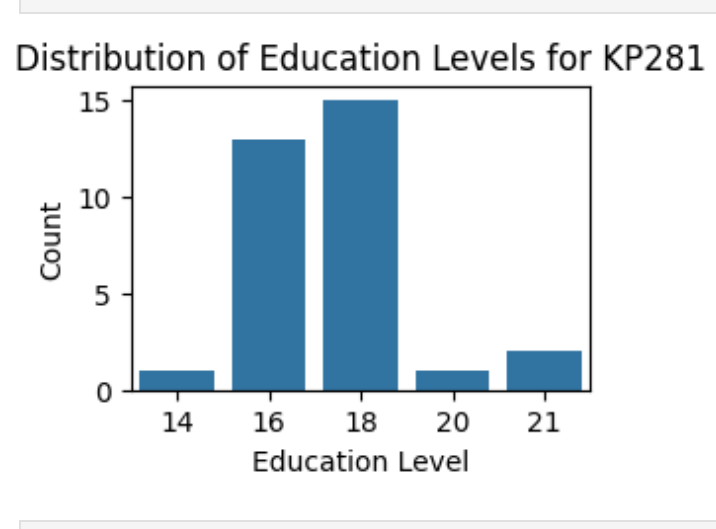
In [178…
```
# Gender: What is the gender distribution of customers?

sns.countplot(third_product_final['Gender'])
```

Out[178…
```
<Axes: xlabel='count', ylabel='Gender'>
```



In [ ]:
```
# Male customers prefer this product more than their counterpart
```

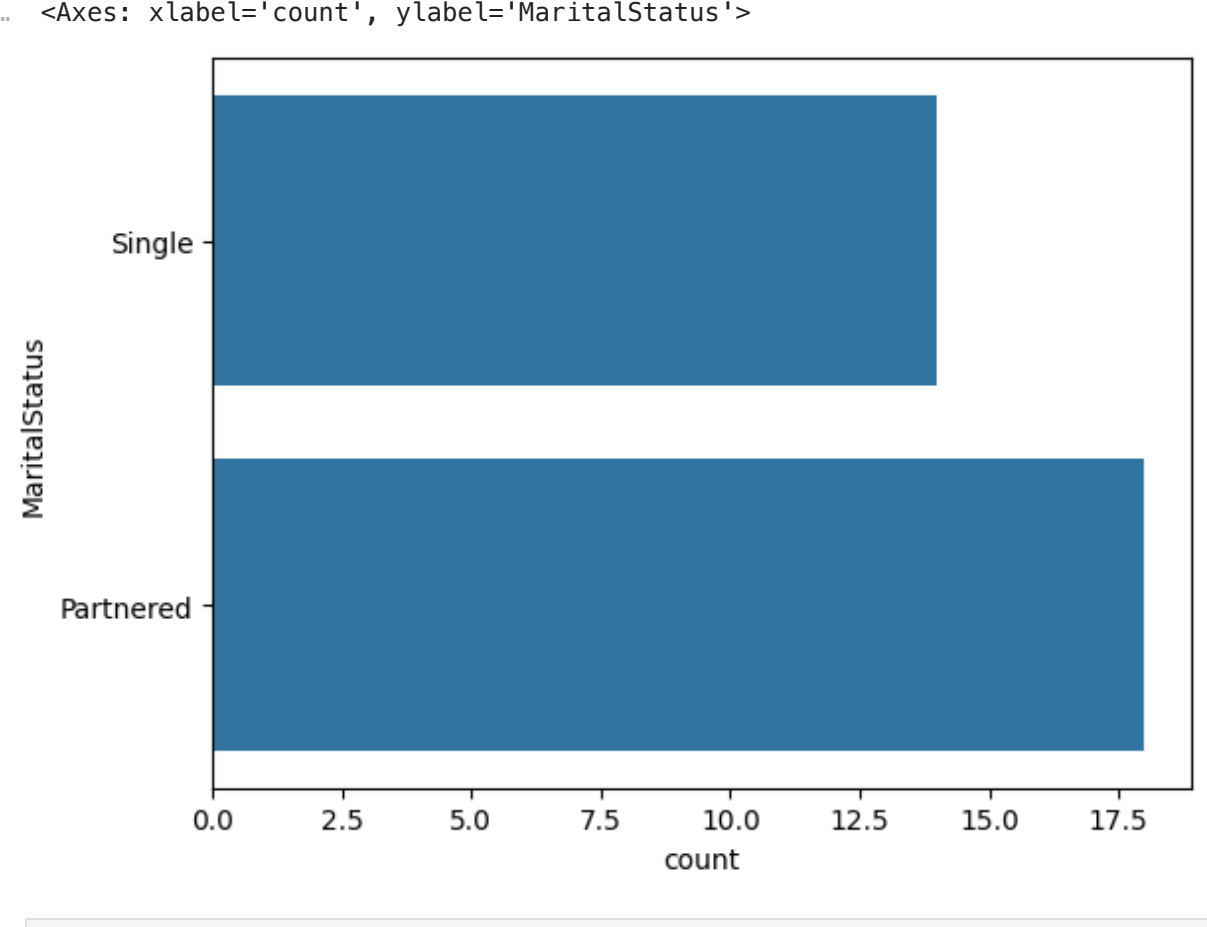
In [179…
```
# Education: What are the common education levels of customers?

education_counts = third_product_final['Education'].value_counts().sort_index()  # Taking discrete values of the Education and then plotting it

plt.figure(figsize=(3, 2))
sns.barplot(x=education_counts.index, y=education_counts.values)
plt.xlabel('Education Level')
plt.ylabel('Count')
plt.title('Distribution of Education Levels for KP281')
plt.show()
```


Distribution of Education Levels for KP281

In [ ]:
```
# the above graphs shows that education grup 18 and 16 are most preferred ones
```

In [180…
```
# MaritalStatus: What is the marital status distribution of customers? Are they mostly single or partnered?

sns.countplot(third_product_final['MaritalStatus'])
```

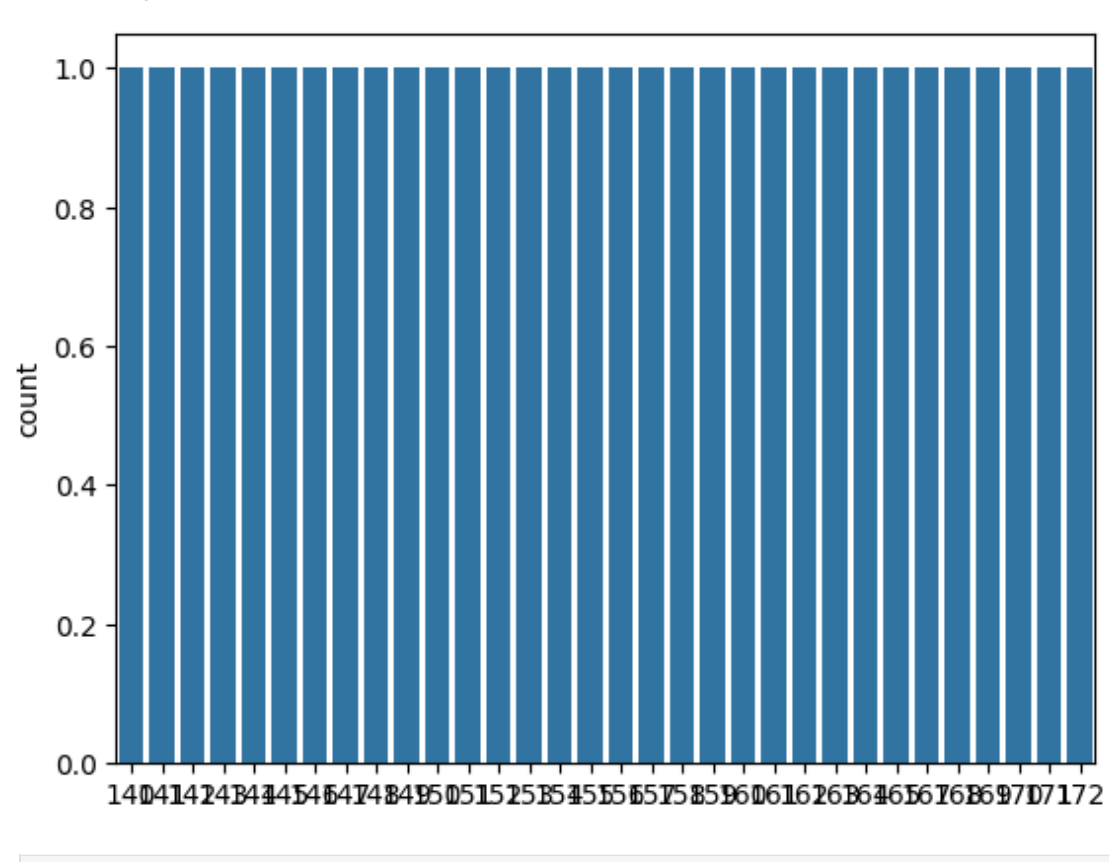
Out[180…
```
<Axes: xlabel='count', ylabel='MaritalStatus'>
```



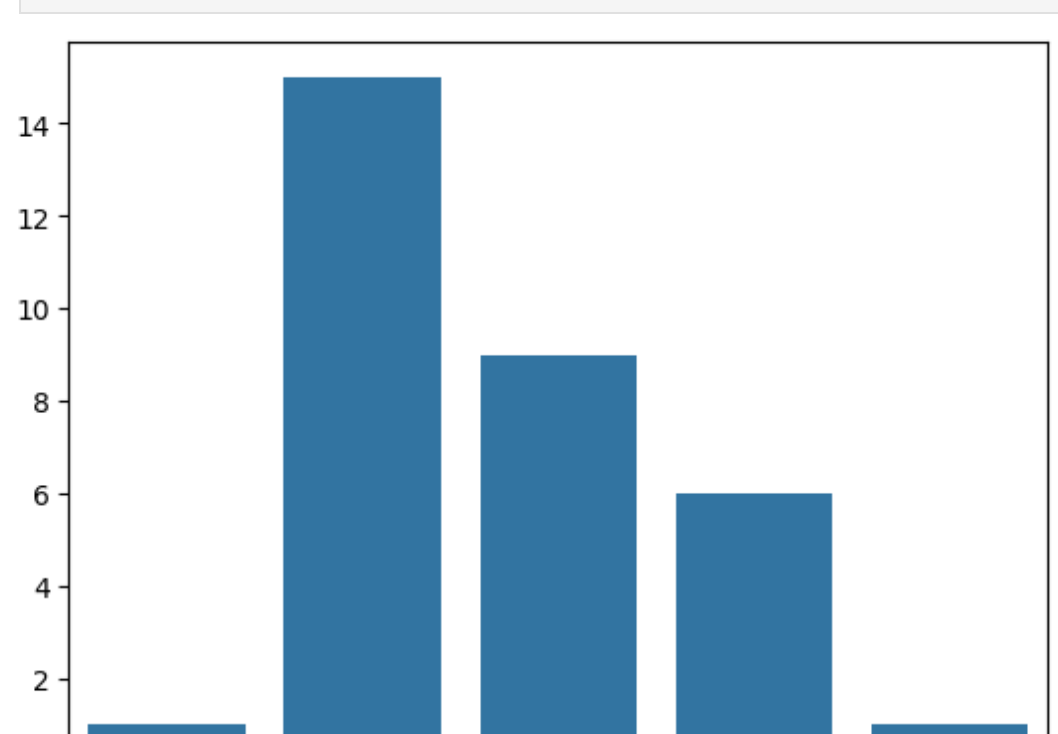
In [ ]:
```
# Partenred i.e. Maried couples are frequent buyer of this product
```

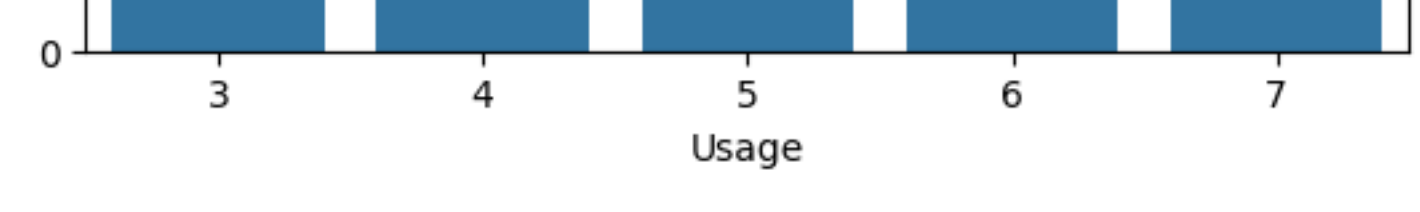
In [181…
```
# Usage: How often do customers plan to use the treadmill per week? What is the most common planned usage?

sns.countplot(third_product_final['Usage'])

# Countplot might be interpreting the numerical data as continous instead of discrete.
# This is causing it to group all values together. Lets calculate the data and use it
# to plot the graph
```

Out[181…
```
<Axes: ylabel='count'>
```



In [182…
```
usage_counts = third_product_final['Usage'].value_counts().sort_index()
usage_counts

sns.barplot(x = usage_counts.index,y=usage_counts.values)
plt.show()
```

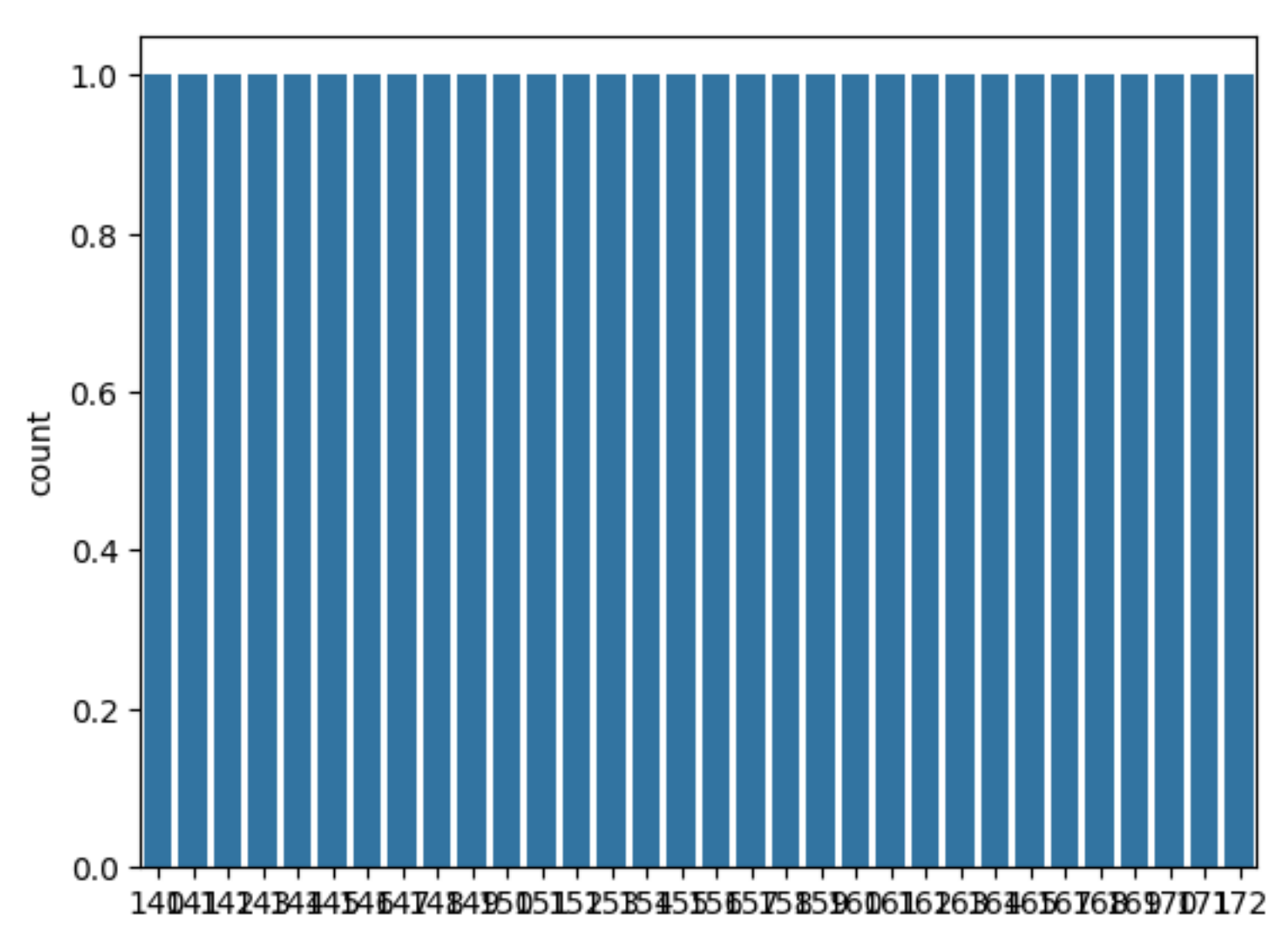




In [ ]:
```
# It is famous amonst the people who wants to use it 4 times in a week
```

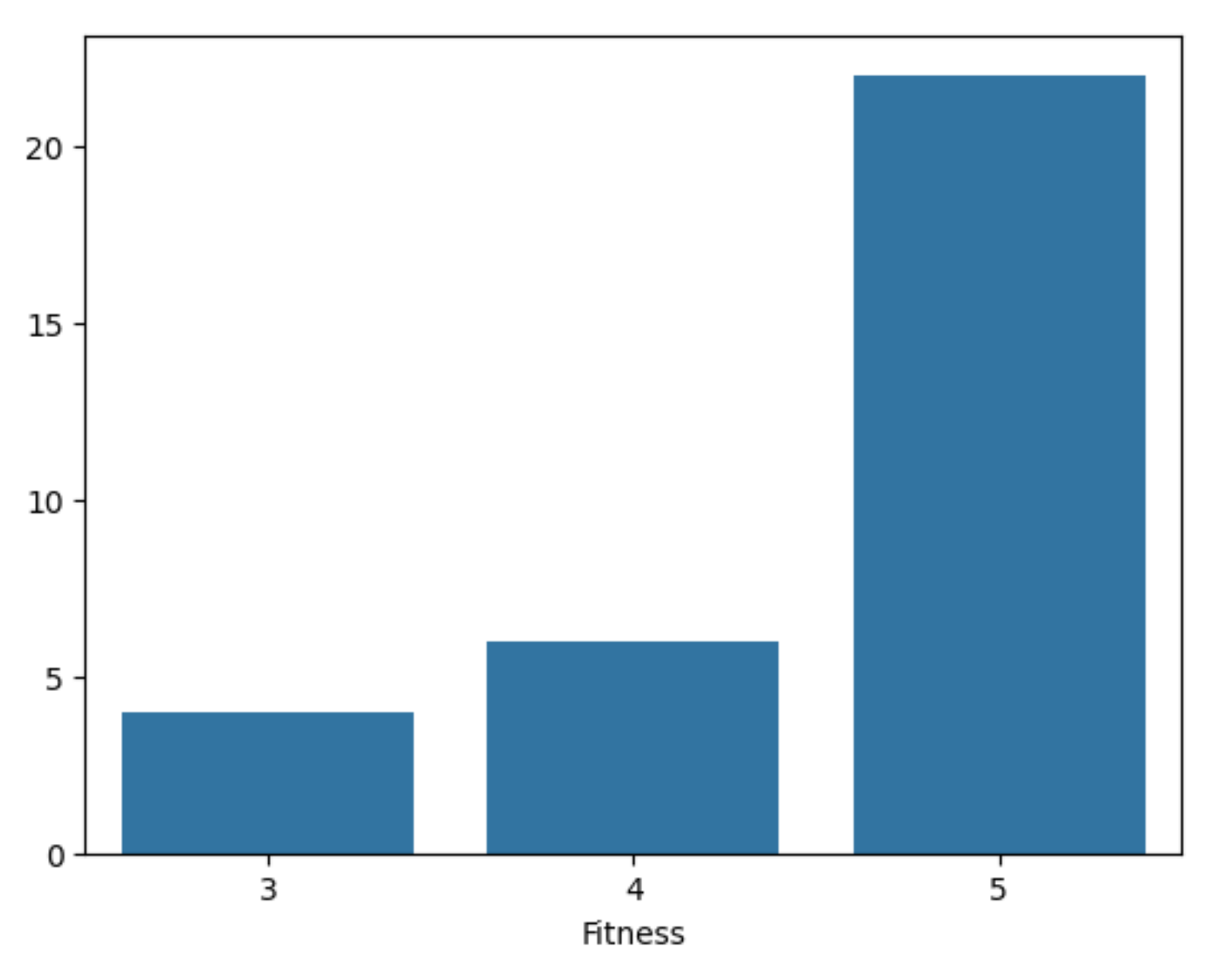
In [183…
```
# Fitness: How do customers rate their own fitness level? What is the average self-rated fitness level?

sns.countplot(third_product_final['Fitness'])

# Repeatig the similar activity to convert continous to discrete
```

Out[183…
```
<Axes: ylabel='count'>
```



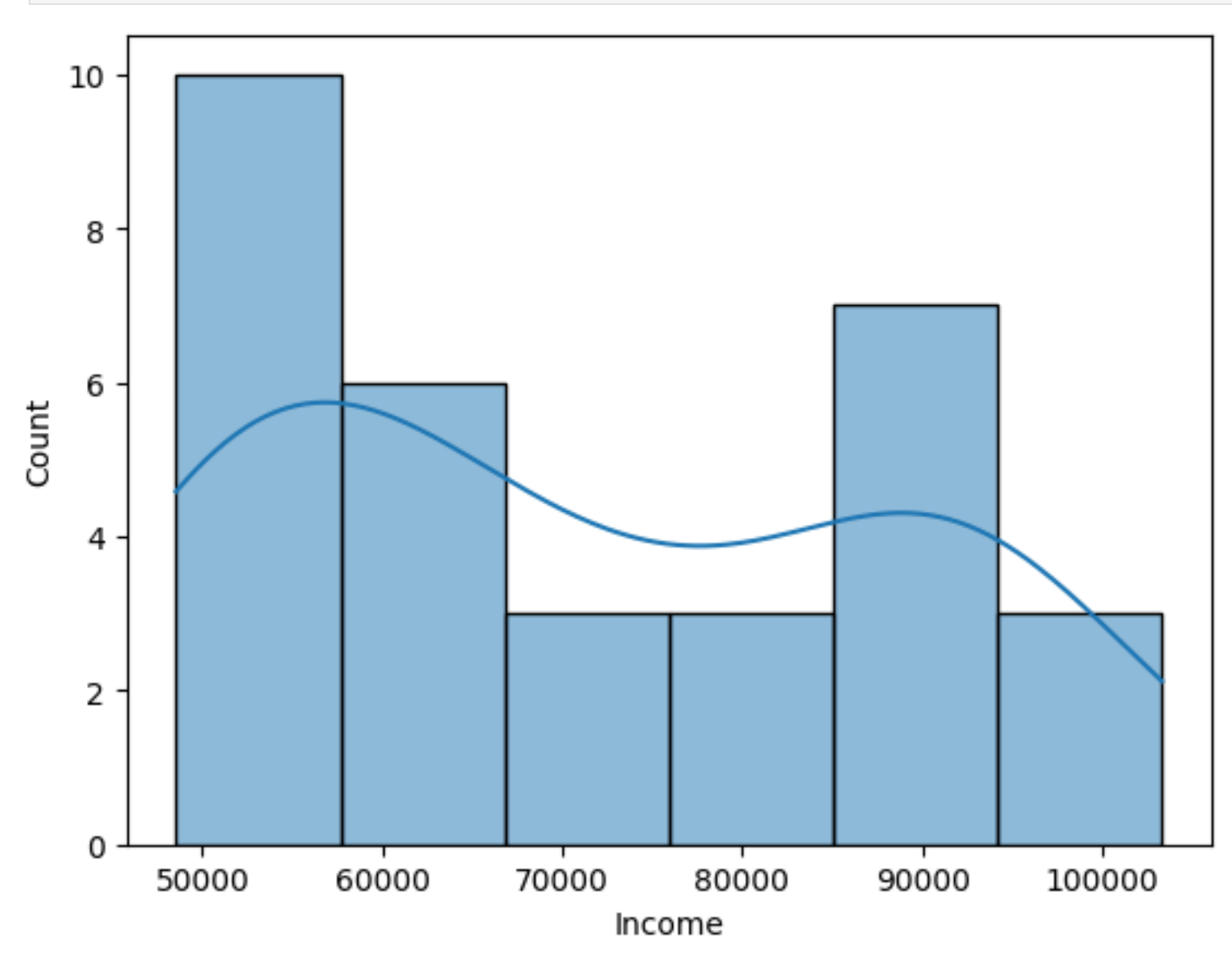
In [184…
```
fitness_counts = third_product_final['Fitness'].value_counts().sort_index()

sns.barplot(x = fitness_counts.index,y=fitness_counts.values)
```

Out[184…
```
<Axes: xlabel='Fitness'>
```



In [ ]:
```
# Customer who are rating themselves 5 are taking this product
```

In [185…
```
# Income: What is the income distribution of customers? What is the average income?

sns.histplot(third_product_final['Income'],kde=True)
plt.show()
```



In [ ]:
```
# Customers whose income is between $ 50000 - $60000 prefers the product and generally this is preferred by high values customer
```

Summarizing the findings till now (Customer Profile for KP781):

Based on the univariate analysis for the KP781 product, the typical customer is between 24 and 26 years old, with a higher preference among male customers. They commonly have education levels of 18 and 16 years and are more likely to be partnered. These customers primarily plan to use the treadmill 4 times per week and rate their fitness level as 5. Their income typically falls between 50,000and60,000, and this product is generally preferred by high-value customers.

Collating the information obtained till now wrt three product:

* KP-281: Based on the univariate analysis for the KP281 product, the typical customer is between 18 and 30 years old, making up about 65% of buyers, with a nearly equal distribution between genders. They commonly have education levels of 14 or 16 years and are more likely to be partnered. These customers primarily plan to use the treadmill 3 times per week and rate their fitness level as 3. Their income typically falls between 45,000and 55,000.

* KP-481: Based on the univariate analysis for the KP481 product, the typical customer is between 18 and 35 years old, making up about 78% of buyers, with a higher preference among female customers. They are more likely to be partnered and primarily plan to use the treadmill 3 times per week. These customers typically rate their fitness level as 3 and have incomes ranging from 45,000to 55,000.

* KP-781: Based on the univariate analysis for the KP781 product, the typical customer is between 24 and 26 years old, with a higher preference among male customers. They commonly have education levels of 18 and 16 years and are more likely to be partnered. These customers primarily plan to use the treadmill 4 times per week and rate their fitness level as 5. Their income typically falls between 50,000and 60,000, and this product is generally preferred by high-value customers.

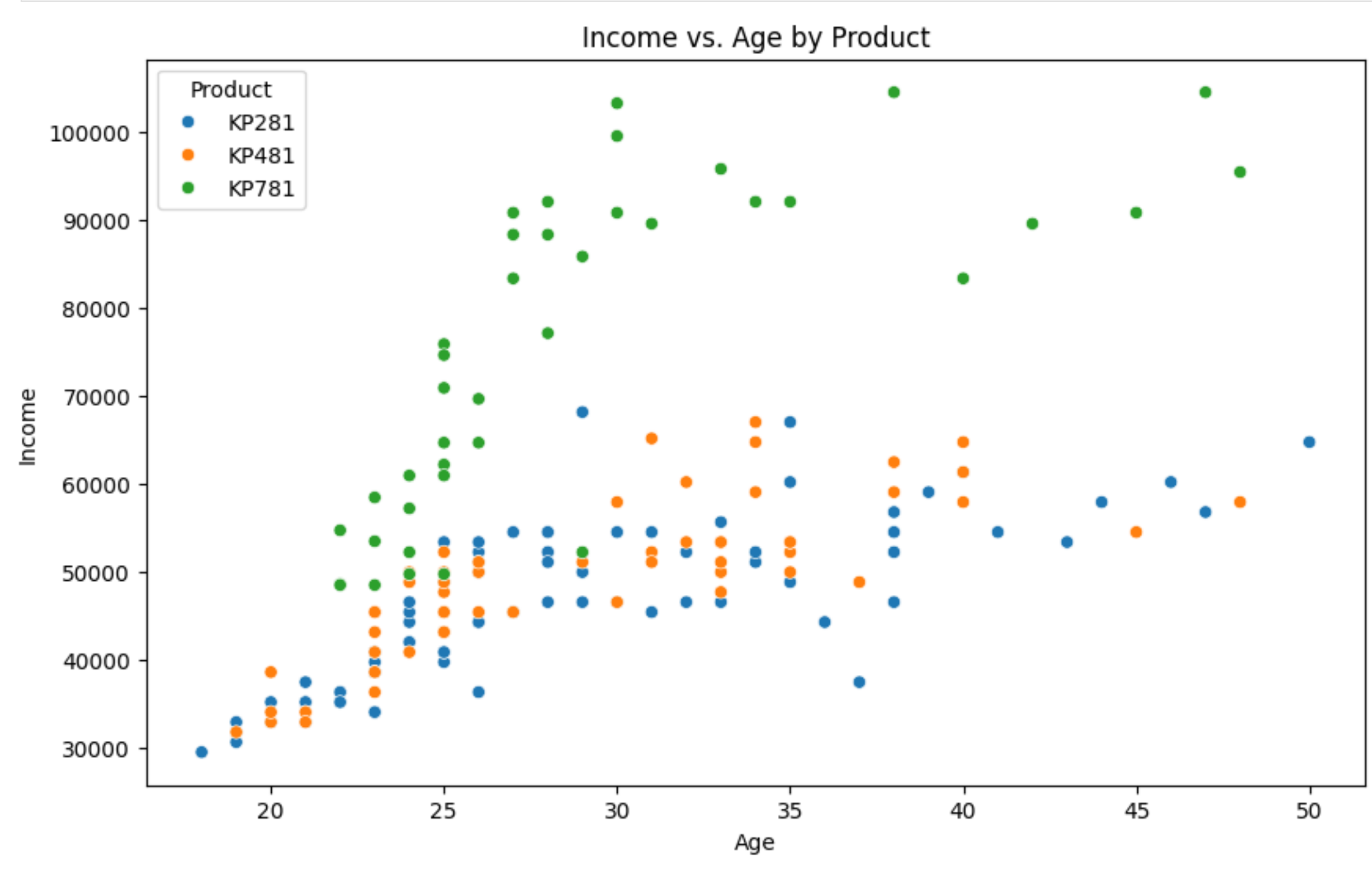
Bi-variate Analysis

In [193…
```
first_product.head()
```

Out[193…

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

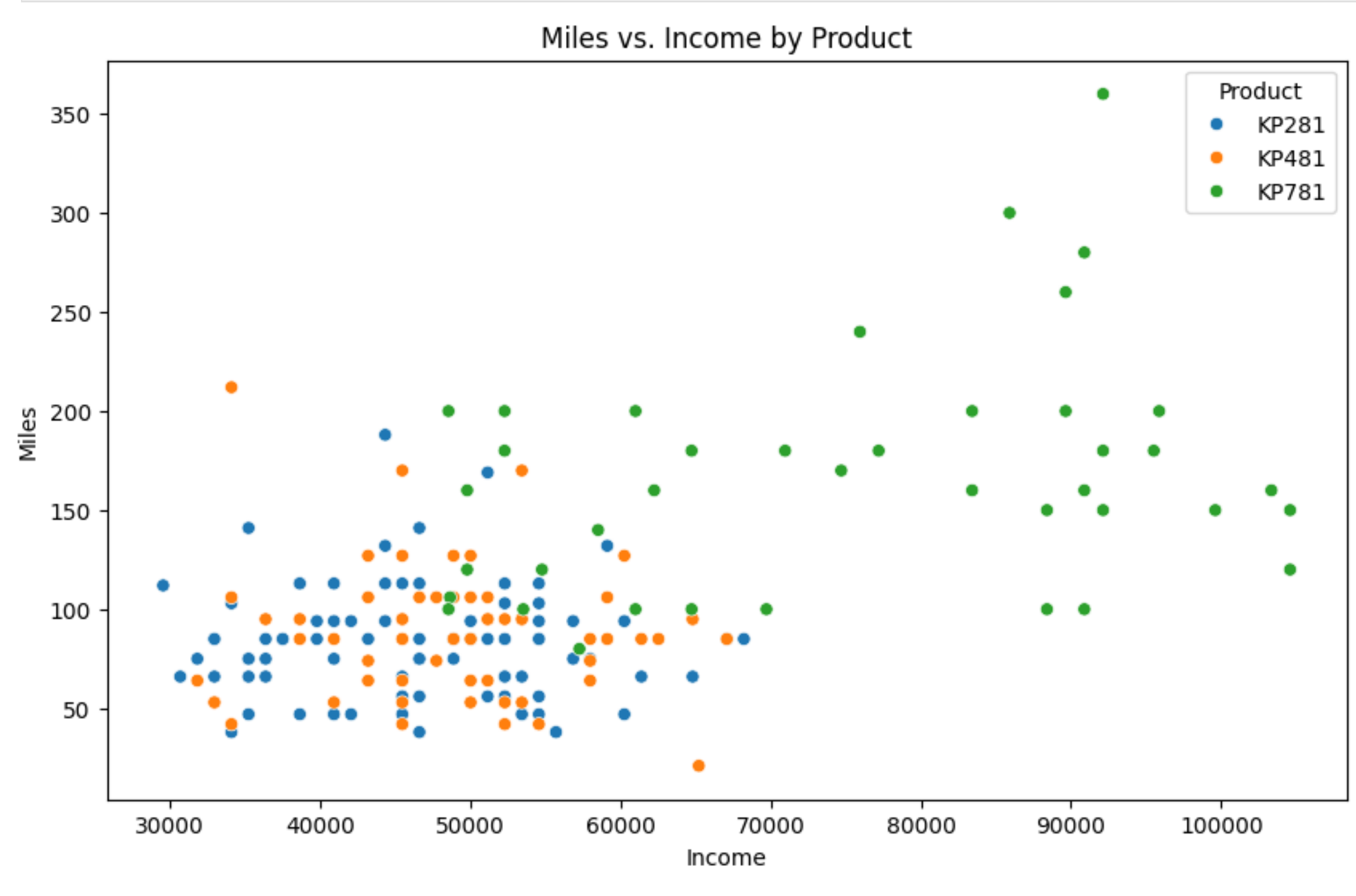
In [199…
```
# Create a scatter plot of Income vs. Age, colored by Product
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='Age', y='Income', hue='Product')
plt.title('Income vs. Age by Product')
plt.xlabel('Age')
plt.ylabel('Income')
plt.show()
```



Inferences:

- The KP281 and KP481 treadmills are more popular among younger to middle-aged individuals with lower to moderate incomes. There isn't a strong differentiation between KP281 and KP481 based on just Age and Income in this plot, suggesting other factors might influence the choice between these two models.
- The KP781 treadmill is clearly targeted towards a different customer segment: older individuals with higher incomes. This suggests it's a premium product purchased by customers with more disposable income.

In [200…
```
# Create a scatter plot of Miles vs. Income, colored by Product
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='Income', y='Miles', hue='Product')
plt.title('Miles vs. Income by Product')
plt.xlabel('Income')
plt.ylabel('Miles')
plt.show()
```

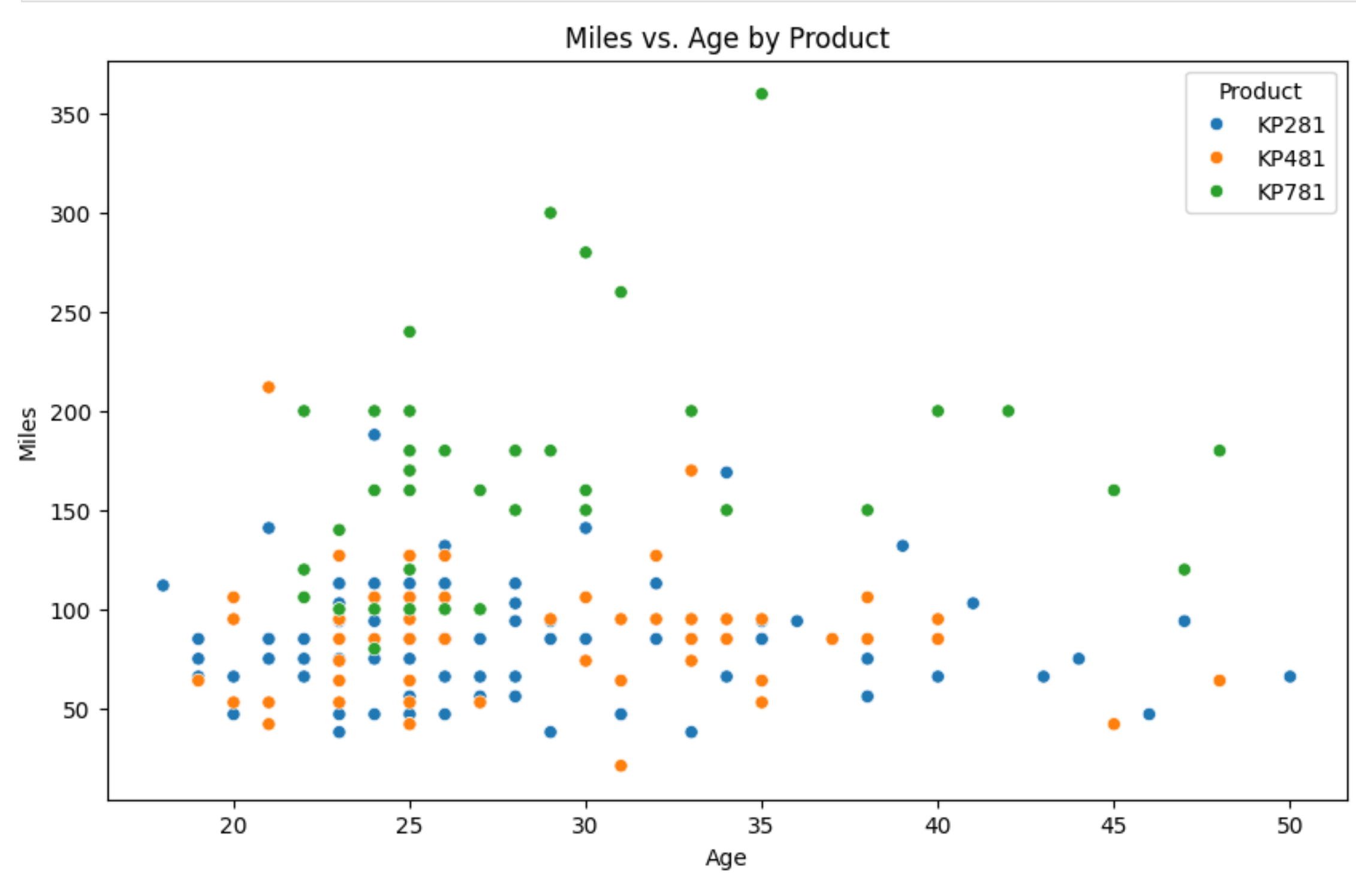


Inferences

- While for the lower-end products (KP281 and KP481), there isn't a very strong relationship between income and miles, the plot clearly shows that customers who purchase the high-end product (KP781) have both higher incomes and higher expected mileage.
- Customers willing to invest in a more expensive treadmill (indicated by higher income) also tend to be those who plan to use it more intensively (indicated by higher miles).
- This reinforces the idea that the KP781 is for more serious users who are likely to be more dedicated to their fitness goals and have the income to support that investment.

In [ ]:

In [201…
```
# Create a scatter plot of Miles vs. Age, colored by Product
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='Age', y='Miles', hue='Product')
plt.title('Miles vs. Age by Product')
plt.xlabel('Age')
plt.ylabel('Miles')
plt.show()
```



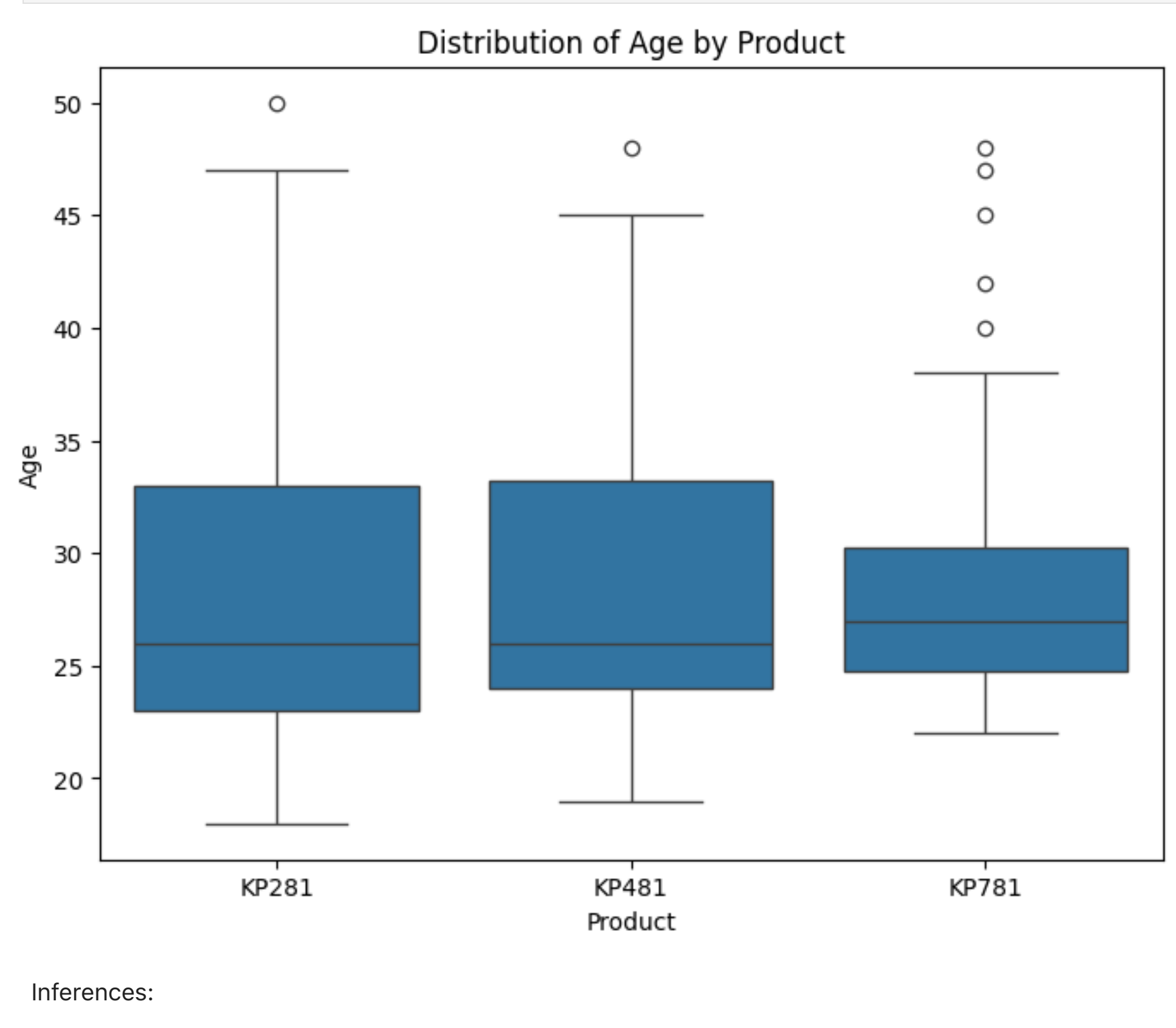
Inferences:

- This plot suggests that for the KP281 and KP481, age doesn't seem to be a strong predictor of expected mileage. Customers of various ages who buy these models generally plan for lower weekly mileage.

- This plot, combined with the Income vs. Age plot, helps build a clearer picture of the distinct customer profile for the KP781 as higher-income individuals who are serious about running, regardless of being younger or middle-aged.

In [ ]:

In [202…
```
# Create box plots of Age by Product
plt.figure(figsize=(8, 6))
sns.boxplot(data=df, x='Product', y='Age')
plt.title('Distribution of Age by Product')
plt.xlabel('Product')
plt.ylabel('Age')
plt.show()
```



Inferences:

- Both the KP281 and KP481 attract a similar age demographic, primarily younger to middle-aged adults. The KP781 also attracts customers from this general age range, but with a slightly higher median age and a tighter concentration, suggesting it appeals more to established professionals within that age bracket.

In [210…
```
# Constructing a two-way contingency table between 'Product' and 'Gender'
ct_gender = pd.crosstab(df['Product'], df['Gender'], margins=True)
print(ct_gender)
```

```
Gender   Female  Male  All
Product
KP281        40    40   80
KP481        29    31   60
KP781         7    33   40
All          76   104  180
```

In [221…
```
# Constructing a two-way contingency table between 'Product' and 'MaritalStatus'
ct_marital_status = pd.crosstab(df['Product'], df['MaritalStatus'], margins=True)
print(ct_marital_status)
```

```
MaritalStatus  Partnered  Single  All
Product
KP281                 48      32   80
KP481                 36      24   60
KP781                 23      17   40
All                  107      73  180
```

In [222…
```
# Probability that partnered customer buy product is:

ct_marital_status.loc['All','Partnered'] / ct_marital_status.loc['All','All']
```

Out[222…
```
np.float64(0.5944444444444444)
```

In [223…
```
# Probability that single customer buy product is:

ct_marital_status.loc['All','Single'] / ct_marital_status.loc['All','All']
```

Out[223…
```
np.float64(0.40555555555555556)
```

In [224…
```
# Probability that customer will buy KP281 given that they are partnered is:

ct_marital_status.loc['KP281','Partnered'] / ct_marital_status.loc['KP281','All']
```

Out[224…
```
np.float64(0.6)
```

In [225…
```
# Probability that customer will buy KP281 given that they are single is:

ct_marital_status.loc['KP281','Single'] / ct_marital_status.loc['KP281','All']
```

Out[225…
```
np.float64(0.4)
```

In [226…
```
# Probability that male customer will buy product is:


ct_gender.loc['All','Male'] / ct_gender.loc['All','All']
```

Out[226…
```
np.float64(0.5777777777777777)
```

In [227…
```
# Probability that male customer will buy KP281 product is:

ct_gender.loc['KP281','Male'] / ct_gender.loc['KP281','All']
```

Out[227…
```
np.float64(0.5)
```

Business Recommendations

Based on the analysis of the AeroFit treadmill customer data, here are some recommendations:

For the KP281 Product:

- **Marketing Focus:** Target marketing campaigns towards younger to middle-aged adults (18-30 years old) with lower to moderate incomes (around $45,000-$55,000). Highlight the product's value and suitability for general fitness and moderate weekly usage (around 3 times per week).
- **Sales Strategy:** Emphasize the product's benefits for partnered individuals, as they are more likely to purchase this model. Sales pitches can focus on shared fitness goals for couples or families.
- **Educational Content:** Since education levels of 14 and 16 years are common, provide clear and easy-to-understand product information and guides.

For the KP481 Product:

- **Marketing Focus:** Similar to KP281, target younger to middle-aged adults (18-35 years old) with lower to moderate incomes ($45,000-$55,000). However, tailor messaging to appeal more to female customers, as they show a higher preference for this model.
- **Sales Strategy:** Continue to emphasize the benefits for partnered individuals.
- **Product Positioning:** Since the customer profile is very similar to KP281 in terms of age, income, and usage frequency, consider how to clearly differentiate the KP481 in marketing to avoid cannibalization and highlight its specific advantages, perhaps focusing on design or slightly different features if applicable.

For the KP781 Product:

- **Marketing Focus:** Target higher-income individuals (around $50,000-$60,000 and above) who are serious about running and fitness. Highlight the product's advanced features, durability, and suitability for higher weekly mileage (around 4+ times per week and higher expected miles).
- **Sales Strategy:** Focus on the value proposition for customers who rate their fitness level as 5 and are looking for a high-performance machine. Emphasize the long-term benefits and capabilities of the product for dedicated runners.
- **Channel Strategy:** Consider marketing and sales channels that reach a more affluent and fitness-focused demographic.

General Recommendations:

- **Marital Status:** Given that partnered individuals are more frequent buyers across all products, explore targeted promotions or product bundles that appeal to couples or families.
- **Fitness Level and Usage:** Tailor marketing messages and product demonstrations to different fitness levels and planned usage frequencies. For example, highlight the ease of use for beginners (Fitness level 3 for KP281/KP481) and the advanced training programs for more serious users (Fitness level 5 for KP781).
- **Further Analysis:** Conduct further analysis to understand the factors that differentiate KP281 and KP481 buyers more clearly, as their profiles are quite similar based on the current analysis. Explore other variables or potential interactions.

In [ ]: